
Robust Sublinear Convergence Rates for Iterative Bregman Projections

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Abstract

1 Entropic regularization provides a simple way to approximate linear programs
2 whose constraints split into two or more tractable blocks. The resulting objec-
3 tives are amenable to cyclic Kullback–Leibler (KL) Bregman projections, with
4 Sinkhorn-type algorithms for optimal transport, matrix scaling, and barycenters as
5 canonical examples. This paper gives a general blueprint for proving $O(1/k)$ dual
6 convergence rate with a constant that scales only *linearly* in $1/\gamma$, where γ is the
7 entropic regularization parameter. We call such rates “robust”, because this mild
8 dependence on γ underpins favorable complexity bounds for approximating the
9 unregularized problem via alternating KL projections. The blueprint reduces the
10 proof to a uniform primal bound and a dual bound for a quotient norm induced by
11 the constraint split. To make these inputs usable, we propose two helper results,
12 which rely on the non-expansiveness of the dual iterations in this quotient dual
13 norm. Instantiating this blueprint for graph-structured transport yields a new flow-
14 Sinkhorn algorithm for the Wasserstein-1 distance on graphs. It achieves ε -additive
15 accuracy on the transshipment cost in $O(p \text{ diameter}^3/\varepsilon^4)$ arithmetic operations
16 (up to logarithmic factors), where p is the number of edges. We also provide
17 a machine-checked Lean formalization of the core blueprint and its graph-W₁
18 instantiation.

19 1 Introduction

20 Iterative Bregman projections power many scalable ML solvers: they turn entropy-regularized con-
21 strained problems into alternating normalization steps implemented with sparse linear algebra and
22 GPU-friendly tensor operations. They underlie Sinkhorn-type methods for optimal transport [Cu-
23 turi, 2013] and their generalizations, such as barycenters [Benamou et al., 2015] and unbalanced-
24 OT [Chizat et al., 2018]. This paper provides a blueprint for proving sublinear convergence with
25 constants that remain “robust” at small regularization. The Wasserstein-1 distance on graphs is one
26 concrete instance of this blueprint: the graph structure changes the split and the dual quotient norm
27 with respect to the vanilla Sinkhorn analysis, but the proof still follows the same route.

28 **Entropic regularization of structured linear programs.** We consider a feasible linear program
29 $\min_{Ax=b, x \geq 0} \langle x, C \rangle$ whose constraints $Ax = b$ naturally split into blocks. We focus on the two-block
30 case for clarity. Throughout, let $d \in \mathbb{N}$ be the dimension of the problem and write $\mathbb{R}_+^d$ for the
31 positive orthant over which the optimization is carried out. Let $A = (A_1; A_2)$, with $A_1 \in \mathbb{R}^{m_1 \times d}$,
32 $A_2 \in \mathbb{R}^{m_2 \times d}$ and $m = m_1 + m_2$, and split $b \in \mathbb{R}^m$ as $b = (b_1; b_2)$ with $b_1 \in \mathbb{R}^{m_1}$ and $b_2 \in \mathbb{R}^{m_2}$.
33 We define the two affine constraint sets

$$C_1 := \{x \in \mathbb{R}_{++}^d : A_1 x = b_1\}, \quad C_2 := \{x \in \mathbb{R}_{++}^d : A_2 x = b_2\}.$$

34 We assume $\mathcal{C}_1 \cap \mathcal{C}_2 \neq \emptyset$, and in particular $b \in \text{range}(A)$. This linear program can thus be re-written
 35 as

$$\min_{x \in \mathbb{R}_+^d} \{\langle C, x \rangle \quad \text{s.t.} \quad A_1 x = b_1, \quad A_2 x = b_2\} = \min_{x \in \mathcal{C}_1 \cap \mathcal{C}_2} \langle C, x \rangle \quad (\mathcal{P}_0)$$

36 for some cost $C \in \mathbb{R}^d$. Fix a reference vector $z \in \mathbb{R}_{++}^d$. For a temperature $\gamma > 0$, we consider the
 37 following entropically regularized problem, which aims to approximate the original feasible linear
 38 program using fast iterative schemes

$$\min_{x \in \mathcal{C}_1 \cap \mathcal{C}_2} \langle C, x \rangle + \gamma \text{KL}(x|z), \quad \text{where} \quad \text{KL}(x|z) := \sum_{i=1}^d \left(x_i \log \frac{x_i}{z_i} - x_i + z_i \right). \quad (\mathcal{P}_\gamma)$$

39 It is common to use a constant reference $z = \alpha \mathbb{1}_d$ for some scale parameter $\alpha \in \mathbb{R}_+$. Since we
 40 do not restrict our attention to probability vectors x (as in optimal transport), the choice of α is
 41 meaningful and should reflect prior knowledge about the expected total mass of the solution. The
 42 regularised objective admits a simple *cyclic Bregman projection* scheme detailed in Section 2: project
 43 x alternately onto $\mathcal{C}_1$ and $\mathcal{C}_2$ in the Kullback–Leibler divergence. This method is attractive when
 44 both projections are explicit; the classical Sinkhorn algorithm for optimal transport is the canonical
 45 example.

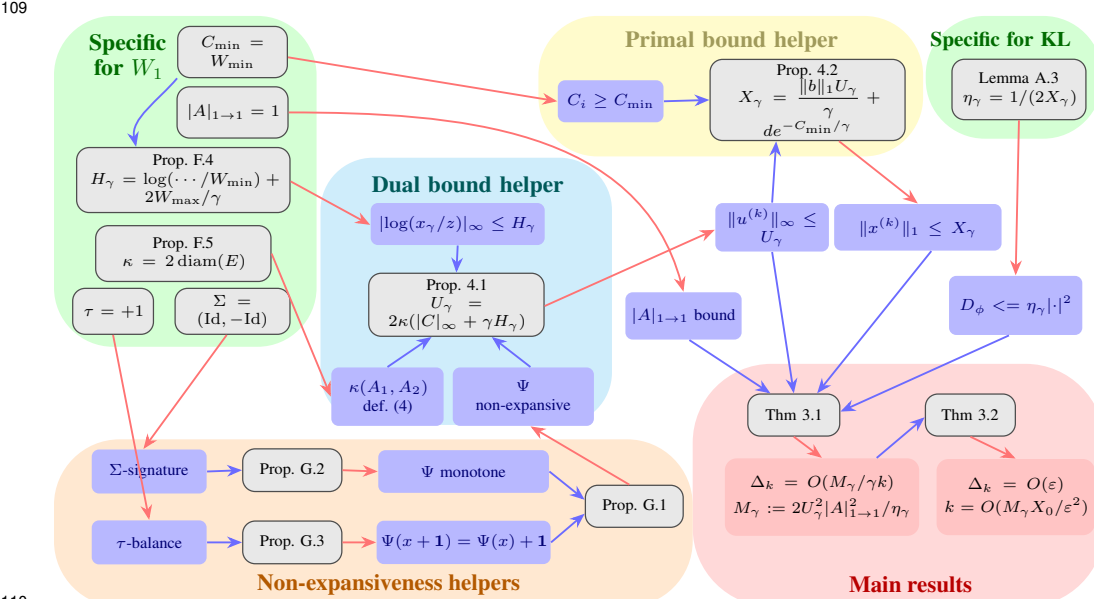
46 This paper establishes sharp sub-linear $O(1/(\gamma k))$ convergence bounds for the *dual objective* of the
 47 KL-projection scheme and demonstrates their impact through a new algorithm for the Wasserstein–1
 48 distance on graphs.

49 **Iterative Bregman projection.** The idea of alternating Bregman projections originates from the
 50 work of Bregman [Bregman, 1967] and of Csiszár–Tusnády [Csiszár and Tusnády, 1984]. When
 51 the distance is squared Euclidean, this procedure reduces to von Neumann’s alternating projections,
 52 whose linear convergence under an angle condition was already established by Friedrichs [Friedrichs,
 53 1938]. More generally, asymptotic linear rates for Bregman projections were shown in [Censor and
 54 Rezač, 2015], again assuming a qualification property that, in Euclidean settings, corresponds to such
 55 an angle condition. This assumption fails for the KL divergence unless all coordinates of the iterates
 56 remain bounded away from zero, something that, in many applications such as entropically regularised
 57 optimal transport, would require a bound of order $1 - e^{-\|C\|_\infty/\gamma}$, which becomes prohibitively small
 58 when γ is small. These earlier convergence results focus on bounding the distance between primal
 59 iterates and the solution, typically in ℓ^1 norm for the KL divergence. Our interest instead lies in
 60 controlling the *dual* value, which provides a weaker guarantee but leads to sharper constants, avoiding
 61 the exponential blow-up in $1/\gamma$ and making the analysis relevant in high-dimensional machine
 62 learning contexts.

63 **Entropic regularisation and Sinkhorn-type methods.** The link between entropy and transport
 64 can be traced back to Schrödinger’s 1931 formulation of the Brownian bridge problem. In the modern
 65 computational setting, Cuturi [Cuturi, 2013] popularised entropic regularisation for large-scale
 66 optimal transport, showing that the resulting smoothed problems bypass the curse of dimensionality
 67 in empirical settings. The associated iterative scaling scheme was introduced independently multiple
 68 times: it appears as early as Yule’s work in 1912 [Yule, 1912] and was later introduced and analyzed by
 69 Sinkhorn [Sinkhorn, 1964], as well as by Deming–Stephan [Deming and Stephan, 1940]. As reviewed
 70 in [Chizat et al., 2025], convergence analyses for these iterations generally fall into two categories.
 71 The first comprises *linear rates* (i.e., exponential with the iteration index k) results with constants
 72 that degrade exponentially badly when γ is small, which we call non-robust rates following [Chizat
 73 et al., 2025]. One approach uses Hilbert’s projective metric to show contraction [Franklin and
 74 Lorenz, 1989, Borwein et al., 1994], with rates of the order $[1 - e^{-\|C\|_\infty/\gamma}]^k$; see [Chen et al., 2016,
 75 Deligiannidis et al., 2024] for continuous-domain extensions and [Eckstein, 2025] for non-compact
 76 domains. Equivalent dependencies arise from convex optimisation proofs [Marino and Gerolin,
 77 2020, Carlier, 2022], which extend to multi-marginal settings [Greco et al., 2023, Conforti et al.,
 78 2023]. For arbitrary, possibly adversarial, cost, this dependence on $\|C\|_\infty/\gamma$ appears tight. The
 79 second category covers polynomial-rate bounds (typically $1/k$) with constants that remain stable as γ
 80 changes (what we call “robust” rates). Early complexity results of this type go back to [Kalantari et al.,
 81 2008] and were sharpened in later works [Altschuler et al., 2017, Chakrabarty and Khanna, 2021,
 82 Dvurechensky et al., 2018, Chizat et al., 2020]. One obtains rate on the dual objective of the order
 83 $\|C\|_\infty^2/(\gamma k)$. This approach can be reframed as mirror descent in a tailored geometry [Léger, 2021,

84 Aubin-Frankowski et al., 2022]. In continuous cases (or in discrete cases if one accepts dependency
 85 on the number of Dirac masses), it is, however, possible to obtain the best of both worlds, and [Chizat
 86 et al., 2025] shows that when the marginals are bounded from below that the linear rate is of the
 87 order $[1 - \kappa\gamma^2/C_{\max}^2]^k$ for some constant κ . Our contributions focus on sublinear rates, and aim at
 88 understanding the general structure that enables robust rates, to apply this analysis to a larger class of
 89 linear programs.

90 **Wasserstein-1 on graphs.** The Wasserstein-1 distance has a particular structure that makes it
 91 both more robust and more tractable than W_2 : it is less sensitive to outliers, and it admits a flow
 92 formulation that is often faster to compute. It has been applied in computer vision [Rubner et al.,
 93 1998, Grauman and Darrell, 2004], machine learning [Kusner et al., 2015], graphics [Solomon et al.,
 94 2014], community detection [Sia et al., 2019], and biology [Sandhu et al., 2015]. When the ground
 95 cost is the shortest-path distance on a graph, W_1 can be formulated as a minimum-cost flow problem
 96 with a fixed divergence constraint [Beckmann, 1952], see for instance [Carlier and Santambrogio,
 97 2012] for applications of this framework to PDEs on continuous domains. Classical solvers such as
 98 the network simplex can be used, and Orlin’s strongly polynomial algorithm achieves $O(pn \log n)$
 99 time on a graph with n vertices and p edges [Ahuja et al., 1993]. The current best exact min-cost flow
 100 algorithm runs in $O(p^{1+o(1)})$ time with high probability [Chen et al., 2025]. For planar graphs with
 101 polynomially bounded integer data, a nearly linear-time exact algorithm is known [Dong et al., 2025],
 102 and for tree metrics (generalizing the 1-D case), W_1 can be computed in linear time $O(n)$ [Evans and
 103 Matsen, 2012]. For approximate solvers, which are the focus of this paper, interior-point methods
 104 combined with fast Laplacian solvers yield additive- ε approximations for min-cost generalized flows
 105 in $O(p^{3/2} \log(1/\varepsilon))$ time [Daich and Spielman, 2008]. In this work, we propose a simpler alternative
 106 based on entropic regularization, with complexity $O(p \text{diameter}(E)^3/\varepsilon^4)$ (up to logarithmic factors).
 107 While this has a worse dependence on ε , it is easy to implement (including on GPUs) and scales
 108 linearly with the number of edges p .



110 Figure 1: Logical map of the convergence blueprint and of its KL/flow-Sinkhorn instantiations. Blue arrows
 111 indicate hypotheses used by a theorem, while red arrows indicate the result statement of that theorem.

112 **Contributions.** The core contribution of this paper is a general blueprint for analyzing the con-
 113 vergence speed of iterative Bregman projection methods summarized in Figure 1, which presents
 114 the architecture of the paper and how the abstract blueprint is instantiated for the graph- W_1 flow-
 115 Sinkhorn algorithm. The main convergence results, shown in red, are Theorem 3.1 and Theorem 3.2.
 116 They rely on primal and dual bounds handled by the helper results in yellow and cyan, including
 117 Proposition 4.2 and Proposition 4.1, and crucially on non-expansiveness of the dual update map Ψ ,
 118 treated by the light-orange helper results in Appendix G. The graph- W_1 instantiation, shown in green,
 119 yields the second core contribution detailed in Section 5: the flow-Sinkhorn algorithm, whose sparse

updates scale linearly with the number of edges and lead to an explicit additive-accuracy complexity guarantee, together with numerical diagnostics on synthetic and genomic-inspired sparse graphs, including GPU line-scaling experiments, that illustrate the regularization/runtime tradeoff in the unregularized-accuracy regime. The implementation, benchmark scripts, and Lean formalization are available at <https://github.com/gpeyre/flow-sinkhorn>. Appendices G and C record auxiliary non-expansiveness results and general Bregman extensions.

2 Iterative KL Projections

We begin by casting any linear program with two affine constraints into an *entropically regularised* form. The resulting objective is minimised by the classical *cyclic KL-projection* (a.k.a. iterative Bregman projection) algorithm, whose dual convergence is analysed in Section 3—culminating in the $O(1/k)$ rate of Theorem 3.1. Throughout the section, we keep the presentation self-contained and focus on two blocks for clarity.

Entropic regularisation and cyclic KL projections. Problems $(\mathcal{P}_\gamma)$ can be conveniently re-written as a Bregman projection problem of the tilted reference z^C (also called Gibbs kernel for OT problems)

$$\min_{x \in \mathcal{C}_1 \cap \mathcal{C}_2} \text{KL}(x|z^C), \quad z_i^C := z_i e^{-C_i/\gamma}. \quad (1)$$

The vector z^C is the Gibbs kernel associated with C , γ and z . Formulation (1) reveals the structure exploited in the sequel: alternating KL projections onto $\mathcal{C}_1$ and $\mathcal{C}_2$ provides a scalable solver whose convergence is quantified in Section 3.

Since the constraints are affine, problem (1) can be solved by iterative Bregman projection using the KL Bregman divergence. Starting from $x^{(0)} = z^C$, it performs the two-step sweep

$$x^{(k+\frac{1}{2})} := \arg \min_{x \in \mathcal{C}_1} \text{KL}(x|x^{(k)}) \quad (P_1), \quad x^{(k+1)} := \arg \min_{x \in \mathcal{C}_2} \text{KL}(x|x^{(k+\frac{1}{2})}) \quad (P_2). \quad (2)$$

When each projection has a closed form, or can be computed with only a few inexpensive inner iterations, (P_1) – (P_2) provide an efficient outer loop. For instance, Appendix E revisits the classical optimal-transport splitting that yields Sinkhorn’s method, while Section 5 introduces a new splitting for the Wasserstein-1 flow formulation on graphs.

Dual problem. We now express the entropic programme $(\mathcal{P}_\gamma)$ in the dual variables by introducing the dual functional F_γ , which is the quantity used throughout the paper to measure convergence speed.

Proposition 2.1 (Dual of $(\mathcal{P}_\gamma)$). *Let $z \in \mathbb{R}_{++}^d$ be fixed and set $Z := \sum_{i=1}^d z_i$. Then the primal and dual values satisfy*

$$\min(P_\gamma) = \max_{u \in \mathbb{R}^m} F_\gamma(u) := \langle b, u \rangle + \gamma Z - \gamma \sum_{i=1}^d z_i \exp\left(\frac{(A^\top u)_i - C_i}{\gamma}\right), \quad (\mathcal{D}_\gamma)$$

where $\min(P_\gamma)$ denotes the value of $(\mathcal{P}_\gamma)$. Moreover, any maximiser u^* and the unique primal minimiser x^* are linked by $x^* = x(u^*)$, where $x(u)_i := z_i \exp(((A^\top u)_i - C_i)/\gamma) = z_i^C \exp((A^\top u)_i/\gamma)$, $i = 1, \dots, d$, and u^* is characterised by the stationarity condition $\nabla F_\gamma(u^*) = 0 \iff A x(u^*) = b$.

Proposition 2.1 allows us to translate the cyclic projections (P_1) – (P_2) into a block-coordinate ascent on F_γ as we detail next. Note that while dual maximizers might not be unique, they are unique up to $\ker(A^\top)$, which is important to take into account in the analysis of the algorithm (for classical OT, this corresponds to translation of the dual potential).

Block-coordinate ascent in the dual. With the split $u = (u_1, u_2) \in \mathbb{R}^{m_1} \times \mathbb{R}^{m_2}$ we write $F_\gamma(u) = F_\gamma(u_1, u_2)$. The two primal KL projections (P_1) – (P_2) are exactly one sweep of block maximisation on F_γ : $u_1^{(k+1)} = \Psi_1(u_2^{(k)})$ and $u_2^{(k+1)} = \Psi_2(u_1^{(k+1)})$, where $\Psi_1(u_2) := \arg \max_{u_1 \in \mathbb{R}^{m_1}} F_\gamma(u_1, u_2)$ and $\Psi_2(u_1) := \arg \max_{u_2 \in \mathbb{R}^{m_2}} F_\gamma(u_1, u_2)$, and the corresponding primal variable is obtained

from Proposition 2.1: $x^{(k)} = x(u_1^{(k)}, u_2^{(k)})$. We define the full sweep map on the u_1 variable as $\Psi := \Psi_1 \circ \Psi_2$ so that $u_1^{(k)} = \Psi^k(u^{(0)})$. For several important splittings, for instance, the classical OT split in Appendix E and the flow split in Section 5, these maps have closed-form solutions, which makes this algorithm practical.

3 Dual convergence

The cyclic projections of Section 2 are equivalent to block-coordinate ascent on the dual objective F_γ introduced in Proposition 2.1. This section proves the two main red-box results summarized in Figure 1: Theorem 3.1, which gives a robust $O(1/(\gamma k))$ dual convergence rate, and Theorem 3.2, which converts this rate into an additive-accuracy guarantee for the unregularized linear program. The point of the blueprint is to make these robust rates modular: once primal and dual boundedness are available, the convergence proof is automatic, and Section 4 explains how the helper results obtain these bounds from non-expansiveness of the sweep map in the relevant quotient dual norm.

Definition 3.1 (block-quotient seminorms). *For $u = (u_1, u_2)$ dual variables, we define $\|u_1\|_{V_1} := \inf_{h \in \ker(A^\top)} \|u_1 + h_1\|_\infty$ and $\|u_2\|_{V_2} := \inf_{h \in \ker(A^\top)} \|u_2 + h_2\|_\infty$. The associated block-quotient dual semi-norm, which depends on the split of constraints (A_1, A_2) , is $\|u\|_V := \max\{\|u_1\|_{V_1}, \|u_2\|_{V_2}\}$. Note that in the special case of classical OT (see Appendix E), this corresponds to the so-called variation semi-norm defined in (25).*

The crucial hypotheses for Theorem 3.1 are bounded primal iterates in ℓ^1 norm and bounded dual iterates in the block-quotient semi-norm $\|\cdot\|_V$. The blueprint is designed to establish these hypotheses through two helpers: a dual bound helper based on non-expansiveness of the sweep map in the quotient norm, and a primal bound helper built on top of that dual control. This is where the split (A_1, A_2) enters the analysis, since it determines the quotient geometry used to measure dual radius.

Theorem 3.1 (Sub-linear dual rate). *Let $\{u^{(k)} = (u_1^{(k)}, u_2^{(k)})\}_{k \geq 0}$ be the dual iterates generated by the two block updates, and let $x^{(k)} := x(u^{(k)})$ be the associated primal iterates. Assume uniform bounds $\sup_k \|u^{(k)}\|_V \leq U_\gamma$ and $\sup_k \|x^{(k)}\|_1 \leq X_\gamma$, where the quotient norm $\|\cdot\|_V$ is defined in Definition 3.1, and denote $\|A\|_{1 \rightarrow 1} := \max_{1 \leq j \leq d} \sum_{i=1}^m |A_{i,j}|$. Define the dual gap $\Delta_k := F_\gamma^* - F_\gamma(u^{(k)})$, $F_\gamma^* = \max_{u \in \mathbb{R}^m} F_\gamma(u)$. Then, for every $k \geq 1$,*

$$0 \leq \Delta_k \leq \frac{8X_\gamma U_\gamma^2 \|A\|_{1 \rightarrow 1}^2}{\gamma} \frac{1}{k}. \quad (3)$$

Proof sketch. The proof keeps the primal and dual viewpoints coupled. First, one half-sweep of block ascent enforces one block constraint exactly, so the remaining residual can be converted into a quantitative increase of F_γ by strong convexity of the KL geometry on the current primal mass shell. This gives a per-step ascent bound in terms of the block residuals, with constants depending only on γ , X_γ , and $\|A\|_{1 \rightarrow 1}$. Second, the dual gap is compared to the same residuals by testing the concave objective along a segment from the current dual variable to an optimal dual representative chosen in the quotient class; this is where the block-quotient radius U_γ enters. Combining the two estimates yields a nonlinear descent recursion $\Delta_k - \Delta_{k+1} \geq \alpha \Delta_k^2$ with $\alpha = \gamma/(8X_\gamma U_\gamma^2 \|A\|_{1 \rightarrow 1}^2)$, and summing the reciprocal gaps gives (3). The complete proof is given in Appendix A.

Numerical complexity of approximating the linear program (P_0) . We now turn the dual gap estimate of Theorem 3.1 into a practical stopping rule for the unregularised programme (P_0) . The next theorem combines the robust dual $O(1/k)$ rate with the standard entropic approximation tradeoff, choosing the temperature γ so that regularization and optimization errors share the target accuracy budget. The resulting guarantee uses only the primal and dual bounds X_γ, U_γ .

Theorem 3.2 (Accuracy versus runtime from primal/dual bounds). *Assume the unregularised LP (P_0) admits an optimal solution x_0^* with $\|x_0^*\|_1 \leq X_0^*$. Run cyclic KL projections at temperature $\gamma = \varepsilon/(2X_0^* \log d)$, yielding iterates $(x^{(k)}, u^{(k)})$, and suppose the uniform bounds $\sup_k \|x^{(k)}\|_1 \leq X_\gamma$ and $\sup_k \|u^{(k)}\|_V \leq U_\gamma$ hold (where $\|\cdot\|_V$ is defined in Definition 3.1). For $\varepsilon > 0$, choose $k := \lceil \frac{1}{\varepsilon^2} 64X_\gamma U_\gamma^2 \|A\|_{1 \rightarrow 1}^2 \max(X_0^*, X_\gamma) \log d \rceil$, $\gamma := \frac{\varepsilon}{2X_0^* \log d}$. Then $|F_0^* - F_\gamma(u^{(k)})| \leq \varepsilon$.*

Proof sketch. The estimate is obtained by separating approximation and optimization errors. The regularization bias compares the unregularized optimum with the entropic optimum evaluated at the same feasible point; since the KL penalty is at most $X_0^* \log d$ on a feasible point with mass X_0^* after normalization against the positive reference, the choice $\gamma = \varepsilon/(2X_0^* \log d)$ spends half of the budget on bias. The remaining half is assigned to the dual optimization gap. Inserting this value of γ into Theorem 3.1 and requiring the displayed lower bound on k makes the robust $O(1/(\gamma k))$ term at most $\varepsilon/2$. The dual value, therefore, approximates the original LP value within ε . The detailed proof is in Appendix B.

Beyond KL. The same blueprint extends to general Bregman divergences under a generalized Pinsker condition. Appendix C states the abstract condition and the resulting rate. This extension covers objectives beyond linear programs, including semidefinite programs; quantum optimal transport is one example where the non-commutative Bregman geometry is natural.

4 Primal and Dual Bound Helpers

Our main result, Theorem 3.1, relies on uniform primal and dual bounds, denoted respectively by $X_{\max}$ and $U_{\max}$. We present here two helps to achieve this, shown in the blueprint of Figure 1 as the cyan and yellow boxes. The complete proofs of both helper results are in Appendix D.

Bounding Dual Iterates using Non-expansiveness of Ψ . The most difficult part is controlling the dual iterations. This section defines the geometric constants that govern this control and provides a generic blueprint that leverages the non-expansiveness of Ψ . It requires a primal bound H_γ (measuring deviation from the reference z in dual coordinates) and a control of the conditioning of the splitting through a decomposition constant κ .

Definition 4.1 (Primal bound H_γ in dual coordinates). *We define $H_\gamma \in [0, +\infty]$ so that the minimizer x_γ of $(\mathcal{P}_\gamma)$ satisfies $\|\log x_\gamma - \log z\|_\infty \leq H_\gamma$.*

Definition 4.2 (Decomposition constant κ). *The decomposition constant $\kappa = \kappa(A_1, A_2)$ is*

$$\kappa(A_1, A_2) := \sup_{y \in \text{range}(A^\top), y \neq 0} \inf \left\{ \frac{\|w_1\|_\infty}{\|y\|_\infty} : \exists w_2 \text{ s.t. } A_1^\top w_1 + A_2^\top w_2 = y \right\} \in [0, +\infty]. \quad (4)$$

Proposition 4.1 (Uniform V_1 -bound for alternating maximization). *We assume Ψ is non-expansive with respect to $\|\cdot\|_{V_1}$: $\|\Psi(a) - \Psi(b)\|_{V_1} \leq \|a - b\|_{V_1}$ for all a, b . Let $\{u_1^{(k)}\}_{k \geq 0}$ be generated by the two block updates, and let $u_\gamma = (u_{\gamma,1}, u_{\gamma,2})$ be any maximizer of (D_γ) with associated primal optimum $x_\gamma = x(u_\gamma)$. Then, for all $k \geq 0$, $\|u_1^{(k)}\|_{V_1} \leq \|u_1^{(0)}\|_{V_1} + 2\kappa(\|C\|_\infty + \gamma H_\gamma)$.*

Proof sketch. The non-expansiveness assumption turns the fixed point of the sweep into an orbit center: every iterate remains within its initial distance from $u_{\gamma,1}$ in the quotient norm. It remains to bound this fixed point. At optimality, the relation $x_\gamma = z \odot \exp((A^\top u_\gamma - C)/\gamma)$ implies that $A^\top u_\gamma$ is uniformly bounded by $\|C\|_\infty + \gamma H_\gamma$ after choosing a suitable representative. The decomposition constant κ converts this bound on $A^\top u_\gamma$ into a bound on the first block potential modulo the kernel. Combining these two steps gives the desired bound. See Appendix D for the complete argument.

Bounding Primal Iterates from Dual Iterates. In many cases, one directly has access to a primal bound X_γ (for instance, in classical OT, $X_\gamma = 1$). If this is not the case, the following proposition shows that it is always possible to derive a bound X_γ from U_γ , with the issue being that it blows as $X_\gamma \sim \|b\|_1 U_\gamma / \gamma$ when $\gamma \rightarrow 0$. The main workload is thus to bound the dual potential in the block quotient semi-norm, which needs to be done on a case-by-case basis and exploit the structure of the split (A_1, A_2) .

Proposition 4.2 (Primal bound from a dual bound). *Let $(u^{(k)})_{k \geq 0}$ be the dual iterates of the two block updates and let $x^{(k)} := x(u^{(k)})$ be the corresponding primal iterates. Assume that the cost is lower bounded in the sense that $C_i \geq C_{\min} \geq 0$ for all $i \in \{1, \dots, d\}$, and assume for simplicity that $u^{(0)} = 0$. Assume moreover a uniform block-radius bound $U_\gamma := \sup_{k \geq 0} \|u^{(k)}\|_V < \infty$. Then for every $k \geq 0$, $\|x^{(k)}\|_1 \leq X_\gamma := \|b\|_1 U_\gamma / \gamma + d e^{-C_{\min}/\gamma}$.*

5 Sinkhorn-flow algorithm for W_1 on graphs

This section presents a new algorithm for approximating W_1 on graphs and demonstrates how to instantiate the general KL-projection blueprint. In Figure 1, this application is the green box, which enables the use of the primal and dual helpers to obtain robust rates. The section ends with numerical diagnostics on synthetic and genomic-inspired sparse graphs. Benchmarks are executed on CPU for reproducibility, and the same benchmark code can also be run on GPU.

W_1 distance on graphs. We consider an undirected graph with a vertex set $V = \{1, \dots, n\}$ and edge-length matrix $W \in \overline{\mathbb{R}}_+^{n \times n}$ ($\overline{\mathbb{R}}_+ := \mathbb{R}_+ \cup \{\infty\}$). An entry $W_{i,j} < \infty$ indicates the presence of the edge (i, j) with length $W_{i,j}$; we impose symmetry $W = W^\top$. Let $E = \{(i, j) \in V^2 : W_{i,j} < \infty\}$ and $p := |E|$ be the edge set and its cardinality. We denote $D \in \mathbb{R}_+^{n \times n}$ the shortest-path matrix, so that $D_{i,j}$ is the geodesic distance between vertices i and j . The graph is assumed to be connected, so that D is finite. As recalled in Appendix E, for any cost matrix C (for instance $C = D$, the geodesic distance), the transport distance W_C is defined by the Kantorovich linear program. We consider two probability vectors b_1, b_2 of size $m_1 = m_2 = n$ and take $C = D$. The resulting Optimal Transport distance $W_D(b_1, b_2)$ is the Wasserstein-1 (W_1) distance, and it enjoys an alternate linear programming formulation that leverages the sparsity of the graph adjacency matrix W , the so-called Beckmann formulation [Beckmann, 1952, Santambrogio, 2015]. We introduce *flow* sparse matrices $f \in \mathbb{F} := \{f \in \mathbb{R}_+^{n \times n} : \forall (i, j) \notin E, f_{i,j} = 0\}$ whose entry $f_{i,j}$ encodes the amount transported from j to i . Note that $\mathbb{F}$ has dimensionality p (the set of edges). Define the discrete divergence operator $\text{div}(f) := f^\top \mathbf{1}_n - f \mathbf{1}_n \in \mathbb{R}^n$. Beckmann’s theorem yields the equivalent linear program

$$W_D(b_1, b_2) = \min_{f \in \mathbb{F}} \langle W, f \rangle : \text{div}(f) = b_1 - b_2. \quad (5)$$

Note that feasible f has non-zero entries only on E ; the true variable dimension is therefore p rather than n^2 . In practice, we store flows as sparse edge lists, and all arithmetic counts in Section 5 are expressed in terms of p .

Constraint splitting and the *flow-Sinkhorn* Algorithm. We duplicate the flow variable so that each affine block is easily projected onto. Write $x := (f, g) \in \mathbb{F}^2$ and $d = 2p$, where we recall $p = |E|$ is the number of finite entries of W and that flows are sparse matrices with p non-zero elements. With these lifted variables, the primal problem (5) can be re-written as

$$W_D(b_1, b_2) = \min_{(f,g) \in \mathcal{C}_1 \cap \mathcal{C}_2} \langle W, f \rangle + \langle W, g \rangle. \quad (6)$$

where the constraints can be written as $\mathcal{C}_1 := \{(f, g) : -f \mathbf{1}_n + g^\top \mathbf{1}_n = b_1 - b_2\}$, $\mathcal{C}_2 := \{(f, g) : f = g\}$. We then consider the entropic regularization ($\mathcal{P}_\gamma$) with a fixed reference vector $z = (z_{i,j})_{i,j}$ supported on the edge $(i, j) \in E$

$$\min_{(f,g) \in \mathcal{C}_1 \cap \mathcal{C}_2} \langle W, f \rangle + \langle W, g \rangle + \gamma \text{KL}(f|z) + \gamma \text{KL}(g|z).$$

Primal KL projections. Our Sinkhorn-flow algorithm is obtained by applying the iterative KL projection to solve this regularized problem. The KL projection on the two constraints can be computed explicitly as stated in the following proposition.

Proposition 5.1 (Closed-form KL projections). *Let $(f, g, h) \in \mathbb{F}^3$, then $\text{Proj}_{\mathcal{C}_1}(h, h) = (\text{diag}(s)h, h \text{diag}(s)^{-1})$, $\text{Proj}_{\mathcal{C}_2}(f, g) = (\sqrt{f \odot g}, \sqrt{f \odot g})$, where the scaling is $s = \phi\left(\frac{b_1 - b_2}{h \mathbf{1}_n}, \frac{h^\top \mathbf{1}_n}{h \mathbf{1}_n}\right) \in \mathbb{R}_{++}^n$, with $\phi(t, u) := \frac{\sqrt{t^2 + 4u} - t}{2}$.*

Flow-Sinkhorn algorithm. We call the cyclic KL scheme of Section 2 applied to the lifted variable $x = (f, g)$ the *flow-Sinkhorn* algorithm. Since $\text{Proj}_{\mathcal{C}_1}$ maps on pairs of equal flows, we track the output of this projection which simplifies the algorithmic description by focussing on a single flow variable and a single dual variable. The flow-Sinkhorn algorithm generates flows $f^{(k)}$ converging to the solution of the entropic regularization of the initial linear program (5).

$$(f^{(k+1)}, f^{(k+1)}) = \text{Proj}_{\mathcal{C}_2} \circ \text{Proj}_{\mathcal{C}_1}(f^{(k)}, f^{(k)}) \quad (7)$$

We denote $v^{(k)} \in \mathbb{R}^n$ the dual variable at iteration k of the algorithm and $s^{(k)} = e^{v^{(k)}/(2\gamma)}$ the associated scaling variable, which satisfies

$$f_{i,j}^{(k)} = z_{i,j}^C e^{\frac{v_i^{(k)} - v_j^{(k)}}{2\gamma}}, \quad f^{(k)} = \text{diag}(s^{(k)}) z^C \text{diag}(1/s^{(k)}), \quad (8)$$

where we set $z_{i,j}^C := z_{i,j} e^{-\frac{W_{i,j}}{\gamma}} \in \mathbb{F}$ (with the convention that $z_{i,j}^C = 0$ when there is no edge, i.e. $W_{i,j} = +\infty$). Proposition 5.2 gives closed forms for iterations (7) over these dual variables. The update of the dual is written in a stable way, so that formulas do not blow up numerically when $\gamma \rightarrow 0$, leveraging the stable implementation of the log-sum-exp operator $\mathcal{L}_\gamma(s) := \gamma \log \sum_j \exp(s_j/\gamma)$, implemented as $\mathcal{L}_\gamma(s - \max s) + \max s$. This stable formula is crucial in practice when targeting small regularization. Note that, in contrast with the usual Sinkhorn algorithm where $z_{i,j}^C > 0$, the Gibbs kernel in our setting may have vanishing entries due to the sparsity of the graph. In this case, the classical Sinkhorn algorithm is no longer guaranteed to converge linearly, because the Hilbert metric may fail to be contractive. We emphasize, however, that our algorithm differs from the classical Sinkhorn method, and our analysis does not rely on linear convergence rates.

Proposition 5.2 (Flow–Sinkhorn update in scaling variables). *One has $v^{(k+1)} = \Psi(v^{(k)})$ where $\Psi = \Psi_1 \circ \Psi_2$ (as defined in the two block updates) can be written as*

$$\Psi(v)_i = \frac{1}{2} v_i + \frac{1}{2} (\alpha_i^+(v) - \alpha_i^-(v)) - \gamma \text{arsinh}(\beta_i), \quad \alpha_i^\pm(v) := \mathcal{L}_\gamma(-w_{i,\cdot} \pm v/2), \quad (9)$$

where $\text{arsinh}(m) := \log(\sqrt{1+m^2} + m)$ and $\beta_i := \frac{b_{1,i} - b_{2,i}}{2} e^{-\frac{\alpha_i^+(v) + \alpha_i^-(v)}{2\gamma}}$.

Convergence Analysis. The following theorem establishes dual convergence for the graph-flow instantiation by plugging the graph-specific helper bounds into the general theorem.

Theorem 5.1 (Sinkhorn–flow complexity). *Sinkhorn–flow achieves an ε -additive approximation of the W_1 distance in $O(p \text{diameter}(E)^3/\varepsilon^4)$ operations, up to logarithmic factors in n , provided that $p = o(1/\log(1/\varepsilon))$.*

Proof sketch. The proof instantiates the green box of Figure 1. The variables are $x = (f, g) \in \mathbb{F}^2$, the two constraint blocks are the divergence constraint $A_1(f, g) = b_1 - b_2$ and the equality constraint $A_2(f, g) = 0$, and the dual variables are $u = (v, U)$, where $A_1(f, g) = f \mathbb{1}_n - g^\top \mathbb{1}_n \in \mathbb{R}^n$ and $A_2(f, g) = f - g \in \mathbb{F}$. Appendix F identifies the quotient norms as variation seminorms and proves the signed non-expansiveness needed by the dual helper: the signature is $\Sigma = \text{diag}(+I_E, -I_E)$, the translation parameter is $\tau = +1$, and the second block update satisfies $\Psi_2(v)_{i,j} = (v_j - v_i)/2$ with $\|\Psi_2(v)\|_{V_2} \leq \|v\|_{V_1}$. The same appendix proves the graph decomposition estimate $\kappa \leq 2 \text{diameter}(E)$ and the explicit bound $|A|_{1 \rightarrow 1} = \|A\|_{1 \rightarrow 1} = 2$ for this split. The dual helper therefore gives $U_\gamma = O(\text{diameter}(E)(W_{\max} + \gamma H_\gamma))$. The graph H_γ estimate and the positive-cost lower bound then feed the primal helper, giving a mass bound $X_\gamma = O(\text{diameter}(E)/\gamma + p e^{-W_{\min}/\gamma})$ for probability inputs. Plugging these U_γ , X_γ , and $|A|_{1 \rightarrow 1}$ values into Theorem 3.2, and taking $\gamma \asymp \varepsilon$, yields $O(\text{diameter}(E)^3/\varepsilon^4)$ iterations up to logarithmic factors. Each sweep uses sparse edge operations, hence costs $O(p)$, which gives the stated arithmetic complexity.

Numerical experiments. The public implementation includes a PyTorch implementation of flow-Sinkhorn together with scripts reproducing these plots. We benchmark in the regime emphasized by our theory: approximation speed toward the *unregularized* solution as a function of wall-clock time and regularization γ , rather than raw per-iteration decay at fixed regularization geometry. The error plotted below is the best-so-far relative Euclidean error on the recovered graph flow, $\min_{s \leq t} \|f^{(s)} - f^*\|_2 / \|f^*\|_2$, where f^* is the unregularized min-cost-flow solution and each method is evaluated through the corresponding graph-flow representation. Flow-Sinkhorn and vanilla Sinkhorn are therefore plotted together on the same axes (solid and dashed, respectively), with a common time horizon. This objective should, however, be interpreted with care for machine-learning practice: in many downstream tasks, keeping a non-vanishing (often larger) regularization can improve stability and generalization, so fastest convergence to the unregularized solution is not always the relevant end goal. The benchmark settings are:

- **Line graph.** Segment graph with nearest-neighbor connectivity, $n = 80$ nodes and two localized endpoint measures on the segment extremities; edge weights are unit lengths.

- **Delaunay sparse graphs.** Planar point cloud with $n = 140$ nodes, Delaunay triangulation connectivity, and Euclidean edge weights; localized source/target masses are sampled on spatially separated regions.
- **Single-cell sparse graphs.** We use the Waddington-OT single-cell RNA-seq data of Schiebinger et al. [2019], which follows mouse embryonic fibroblast reprogramming toward pluripotency. The displayed graph is built on a subset of $n = 240$ cells, sampled as 60 cells from each of the first four snapshots (days 0, 0.5, 1, 1.5), embedded by PCA (dimension 30) and connected by a k -NN graph ($k = 4$). The blue and red measures are the empirical distributions at the initial and final sampled snapshots; the displayed flow is therefore a coarse transport proxy for developmental trajectories and fate progression across the sampled cell-state manifold.

For graph visualizations, we aggregate directed flows into undirected magnitudes $|f_{ij}| + |f_{ji}|$, draw all graph edges in thin gray, and overlay only upper-quantile transport edges in thicker orange; source/target supports are shown as blue/red markers. Figures 2 and 3 summarize the CPU runs (for the 3 settings) and the GPU (NVIDIA Quadro RTX 4000 (8GB)) runs (for the line graph only, but with 3 different sizes n). In the GPU runs, increasing the line graph size makes the separation between vanilla Sinkhorn and flow-Sinkhorn more pronounced: the sparse flow formulation keeps its edge-local arithmetic, while vanilla Sinkhorn still acts on the dense path metric.

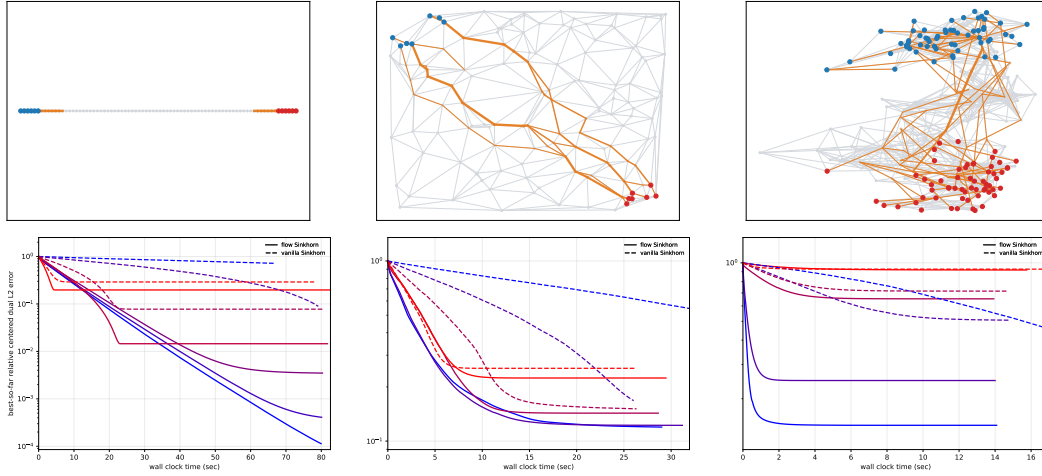


Figure 2: Top row: benched graph together with the flow f (line graph, planar Delaunay, single cell). Bottom row: ℓ^2 error $\|f^{(s)} - f^*\|_2 / \|f^*\|_2$, against wall-clock time. Flow Sinkhorn is plain and vanilla Sinkhorn is dashed; colors encode γ from blue (small) to red (large). Gamma ranges are: line (flow $\gamma \in [10^{-3}, 5]$, vanilla $\gamma \in [10^{-1}, 1]$), Delaunay (flow $\gamma \in [9 \times 10^{-4}, 9 \times 10^{-3}]$, vanilla $\gamma \in [9 \times 10^{-4}, 9 \times 10^{-3}]$), single-cell (flow $\gamma \in [3 \times 10^{-1}, 9]$, vanilla $\gamma \in [1, 5]$).

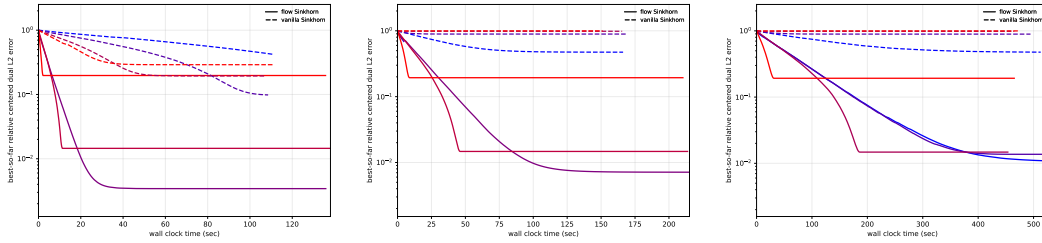


Figure 3: GPU line-graph benchmark, same best-so-far relative ℓ^2 flow error as in Figure 2. From left to right: small ($n = 80, m = 6$), large ($n = 160, m = 12$), and larger ($n = 320, m = 24$) line-segment tests.

Conclusion

We have presented an analysis of iterative Bregman projection methods that yields sublinear convergence rates but with constants that scale favorably with the dimension. Such a regime is the relevant one for studying approximation rates in machine learning, where problems are typically obtained through sampling and discretization, and dimensional dependence plays a central role. While the behavior of these methods was well understood in the classical Optimal Transport setting, it was less clear which structural properties were truly responsible for their stability and effectiveness. This article identifies and isolates these pivotal ingredients, and proposes a generic blueprint that applies beyond the classical case. The relevance of the analysis is illustrated through its application to network flow formulations, leading to a cheap and easy-to-implement algorithm for computing the Wasserstein-1 distance on graphs.

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575 The informal mathematical proofs were completed and carefully checked by the authors;

576 the Lean proofs are provided in the public repository as an additional machine-checkable

577 certificate.

578 A Proof of the sublinear dual rate

579 Proof of Theorem 3.1.

580 *Proof.* The proof relies on a per-step ascent estimate (Lemma A.1) and a dual-gap-to-residual
 581 comparison (Lemma A.2). Write the global residual as $r^{(k)} = Ax^{(k)} - b$ and recall $X_\gamma \geq \|x^{(k)}\|_1$ for
 582 all k . Summing (A1) and (A2) in Lemma A.1 yields, for a full outer sweep, denoting $\lambda := \frac{\gamma}{2X_\gamma \|A\|_{1 \rightarrow 1}^2}$

$$\Delta_k - \Delta_{k+1} \geq \lambda(\|r_1^{(k)}\|_1^2 + \|r_2^{(k+\frac{1}{2})}\|_1^2) \geq \lambda\|r_1^{(k)}\|_1^2.$$

583 By Lemma A.2, $\Delta_k \leq 2U_\gamma \|r_1^{(k)}\|_1$, hence $\|r_1^{(k)}\|_1 \geq \Delta_k/(2U_\gamma)$. Substituting into the previous
 584 inequality gives

$$\Delta_k - \Delta_{k+1} \geq \alpha \Delta_k^2, \quad \text{where} \quad \alpha := \frac{\lambda}{4U_\gamma^2}.$$

585 Dividing by $\Delta_k \Delta_{k+1}$ and using the fact that Δ_k is decaying gives $\frac{1}{\Delta_{k+1}} - \frac{1}{\Delta_k} \geq \alpha$. Summing from
 586 0 to $k-1$ yields

$$\frac{1}{\Delta_k} \geq \frac{1}{\Delta_0} + \alpha k \geq \alpha k,$$

587 since $\Delta_0 > 0$. Therefore, for every $k \geq 1$, $\Delta_k \leq \frac{1}{\alpha k}$, which is exactly the estimate of Theorem 3.1.
 588 $\square$

589 The proof of Theorem 3.1 rests on two ingredients: Lemma A.1 quantifies the ascent achieved during
 590 one half-sweep of the dual ascent the two block updates; Lemma A.2 relates the resulting dual gap
 591 to the primal residuals. In the following, let $u^{(k)}$ denote the dual iterates and $x^{(k)} := x(u^{(k)})$ the
 592 corresponding primal iterates.

593 **Lemma A.1** (Per-step ascent for the dual blocks). *Denoting $r^{(k)} := Ax^{(k)} - b \in \mathbb{R}^m$, assuming*
 594 *uniform bounds on the primal mass $\sup_k \|x^{(k)}\|_1 \leq X_\gamma$, then for every $k \geq 0$ the block updates*
 595 *satisfy*

$$F_\gamma(u_1^{(k+1)}, u_2^{(k)}) - F_\gamma(u_1^{(k)}, u_2^{(k)}) \geq \frac{\gamma}{2X_\gamma} \frac{\|r_1^{(k)}\|_1^2}{\|A\|_{1 \rightarrow 1}^2}, \quad (\text{A1})$$

$$F_\gamma(u_1^{(k+1)}, u_2^{(k+1)}) - F_\gamma(u_1^{(k+\frac{1}{2})}, u_2^{(k)}) \geq \frac{\gamma}{2X_\gamma} \frac{\|r_2^{(k+1)}\|_1^2}{\|A\|_{1 \rightarrow 1}^2}. \quad (\text{A2})$$

596 *Proof.* We prove (A1); the argument for (A2) is identical with the roles of (A_1, u_1, r_1) and
 597 (A_2, u_2, r_2) swapped. Let $\varphi(x) = \sum_i x_i (\log x_i - 1)$, so $D_\varphi(p, q) = \text{KL}(p||q)$. Because $x^{(k+\frac{1}{2})}$ is
 598 the KL-projection of $x^{(k)}$ onto $\mathcal{C}_1 = \{x : A_1 x = b_1\}$, for every $s \in \mathcal{C}_1$,

$$\text{KL}(s|x^{(k)}) = \text{KL}(s|x^{(k+\frac{1}{2})}) + \text{KL}(x^{(k+\frac{1}{2})}|x^{(k)}).$$

599 With $x^{(k)} = x(u_1^{(k)}, u_2^{(k)})$ and $x^{(k+\frac{1}{2})} = x(u_1^{(k+1)}, u_2^{(k)})$, and using $A_1 x^{(k+\frac{1}{2})} = b_1$ and $A_1 x^{(k)} =$
 600 $b_1 - r_1^{(k)}$, a direct calculation shows

$$F_\gamma(u_1^{(k+\frac{1}{2})}, u_2^{(k)}) - F_\gamma(u_1^{(k)}, u_2^{(k)}) = \gamma \text{KL}(x^{(k+\frac{1}{2})}|x^{(k)}).$$

601 By the non-normalised Pinsker inequality, stated for completeness as Lemma A.3 below and applied
 602 on the common mass shell of the KL projection, $\text{KL}(p||q) \geq \frac{\|p-q\|_1^2}{2 \sum_i p_i}$. Taking $p = x^{(k+\frac{1}{2})}$ and using
 603 $\sum_i x_i^{(k+\frac{1}{2})} \leq X_\gamma$ gives

$$\text{KL}(x^{(k+\frac{1}{2})}|x^{(k)}) \geq \frac{\|x^{(k+\frac{1}{2})} - x^{(k)}\|_1^2}{2X_\gamma}.$$

604 Since $r_1^{(k)} = A_1(x^{(k)} - x^{(k+\frac{1}{2})})$,

$$\|r_1^{(k)}\|_1 \leq \|A_1\|_{1 \rightarrow 1} \|x^{(k)} - x^{(k+\frac{1}{2})}\|_1 \leq \|A\|_{1 \rightarrow 1} \|x^{(k)} - x^{(k+\frac{1}{2})}\|_1,$$

605 where we used that $\|A_1\|_{1 \rightarrow 1} \leq \|A\|_{1 \rightarrow 1}$. Combining the previous displays yields (A1). $\square$

606 If the dual iterates stay bounded, the following result relates the current dual gap to the ℓ^1 residuals.
 607 **Lemma A.2** (Dual gap versus global residual). *Let $u^{(k)}$ be the dual iterates and $x^{(k)} := x(u^{(k)})$*
 608 *the corresponding primal iterates. Assume the dual iterates are uniformly bounded in the block-*
 609 *quotient seminorm (defined in Definition 3.1) $\sup_k \|u^{(k)}\|_V \leq U_\gamma$. Let $u^* \in \arg \max F_\gamma$ and write*
 610 $\Delta_k := F_\gamma(u^*) - F_\gamma(u^{(k)})$ *and $r^{(k)} := Ax^{(k)} - b$. Then for every $k \geq 0$,*

$$\Delta_k \leq 2U_\gamma \|r_1^{(k)}\|_1, \quad \Delta_{k+1/2} \leq 2U_\gamma \|r_2^{(k+1/2)}\|_1, \quad (10)$$

611 *Proof.* Concavity and smoothness give, for any dual maximiser u^* ,

$$\Delta_k = F_\gamma(u^*) - F_\gamma(u^{(k)}) \leq \langle \nabla F_\gamma(u^{(k)}), u^* - u^{(k)} \rangle = \langle -r^{(k)}, u^* - u^{(k)} \rangle.$$

612 At the beginning of a sweep, the previous projection enforces $A_2 x^{(k)} = b_2$, hence $r_2^{(k)} = 0$. Therefore

$$\Delta_k \leq \langle -r_1^{(k)}, u_1^* - u_1^{(k)} \rangle.$$

613 Let $h^*, h^{(k)} \in \ker(A^\top)$ be arbitrary. Since $r^{(k)} = [r_1^{(k)}; 0] \in \text{im}(A)$ and $\ker(A^\top) \perp \text{im}(A)$, we
 614 have $\langle r^{(k)}, h^* \rangle = 0$, i.e. $\langle r_1^{(k)}, h_1^* \rangle = 0$; similarly $\langle r_1^{(k)}, h_1^{(k)} \rangle = 0$. Thus

$$\Delta_k = \langle -r_1^{(k)}, (u_1^* + h_1^*) - (u_1^{(k)} + h_1^{(k)}) \rangle \leq \|r_1^{(k)}\|_1 (\|u_1^* + h_1^*\|_\infty + \|u_1^{(k)} + h_1^{(k)}\|_\infty),$$

615 by Hölder. Taking the infimum independently over $h^*, h^{(k)} \in \ker(A^\top)$ yields

$$\Delta_k \leq \|r_1^{(k)}\|_1 (\|u_1^*\|_{V_1} + \|u_1^{(k)}\|_{V_1}) \leq 2U_\gamma \|r_1^{(k)}\|_1,$$

616 because $\|u^{(k)}\|_V \leq U_\gamma$ implies $\|u_1^{(k)}\|_{V_1} \leq U_\gamma$, and dual maximisers share the same V -seminorm
 617 limit so $\|u^*\|_V \leq U_\gamma$. The second inequality is proved similarly using $\|\cdot\|_{V_2}$ in place of $\|\cdot\|_{V_1}$. $\square$

618 **Proposition A.1** (Normalized Pinsker inequality). *For $\mu, \nu \in \Delta_d$,*

$$\text{KL}(\mu|\nu) \geq \frac{1}{2} \|\mu - \nu\|_1^2.$$

619 *Proof.* Using the variational representation of relative entropy,

$$\text{KL}(\mu|\nu) = \sup_{f \in \mathbb{R}^d} \left\{ \langle \mu, f \rangle - \log \left(\sum_i \nu_i e^{f_i} \right) \right\},$$

620 choose $f_i = \lambda s_i$ where $s_i = \text{sign}(\mu_i - \nu_i) \in \{-1, 1\}$ and $\lambda \geq 0$. Then $\langle \mu - \nu, s \rangle = \|\mu - \nu\|_1$.
 621 Writing $a = \sum_{i: s_i=1} \nu_i$, $b = \sum_{i: s_i=-1} \nu_i$ gives $\sum_i \nu_i e^{\lambda s_i} = ae^\lambda + be^{-\lambda}$ with $a + b = 1$, hence

$$\log \left(\sum_i \nu_i e^{\lambda s_i} \right) \leq \lambda \sum_i \nu_i s_i + \frac{\lambda^2}{2}.$$

622 Therefore

$$\text{KL}(\mu|\nu) \geq \lambda \|\mu - \nu\|_1 - \frac{\lambda^2}{2} \quad (\lambda \geq 0),$$

623 and choosing $\lambda = \|\mu - \nu\|_1$ yields the claim. $\square$

624 **Lemma A.3** (Non-normalised Pinsker inequality). *Let $p, q \in \mathbb{R}_+^d$ have the same positive mass*
 625 $M = \sum_i p_i = \sum_i q_i > 0$. *Then*

$$\text{KL}(p|q) \geq \frac{\|p - q\|_1^2}{2M}.$$

626 *Proof.* Set $\bar{p} = p/M$ and $\bar{q} = q/M$. Then $\bar{p}, \bar{q} \in \Delta_d$, so $\text{KL}(\bar{p}|\bar{q}) \geq \frac{1}{2} \|\bar{p} - \bar{q}\|_1^2$ by Proposition A.1.
 627 By homogeneity on a common mass shell, $\text{KL}(p|q) = M \text{KL}(\bar{p}|\bar{q})$ and $\|\bar{p} - \bar{q}\|_1 = M^{-1} \|p - q\|_1$.
 628 Therefore

$$\text{KL}(p|q) = M \text{KL}(\bar{p}|\bar{q}) \geq \frac{M}{2} \|\bar{p} - \bar{q}\|_1^2 = \frac{\|p - q\|_1^2}{2M},$$

629 as claimed. $\square$

B KL bias and runtime proof

Lemma B.1 (KL bias). *Let $F_0^* := \min\{\langle C, x \rangle : x \in \mathbb{R}_{++}^d, Ax = b\}$ be the value of $(\mathcal{P}_0)$ and let F_γ^* be the value of $(\mathcal{P}_\gamma)$. Assume there exists an optimal (unregularised) solution x_0^* of $(\mathcal{P}_0)$ with $\|x_0^*\|_1 \leq X_0^*$. Then*

$$0 \leq F_\gamma^* - F_0^* \leq \gamma \text{KL}(x_0^*|z). \quad (11)$$

For simplicity, we now restrict attention to constant reference vectors $z \propto \alpha \mathbf{1}$, assuming $d \geq 3$ then

$$0 \leq F_\gamma^* - F_0^* \leq \gamma X_0^* \log d \quad \text{when using } z = (X_0^*/d)\mathbf{1}. \quad (12)$$

Proof. Since the KL-regularised objective is $\langle C, x \rangle + \gamma \text{KL}(x|z)$ and $\text{KL}(x|z) \geq 0$ for all $x \in \mathbb{R}_{++}^d$, we have

$$F_\gamma^* = \min_{\substack{x \in \mathbb{R}_{++}^d \\ Ax=b}} (\langle C, x \rangle + \gamma \text{KL}(x|z)) \geq \min_{\substack{x \in \mathbb{R}_{++}^d \\ Ax=b}} \langle C, x \rangle = F_0^*,$$

which gives $0 \leq F_\gamma^* - F_0^*$. Moreover, since x_0^* is feasible for the regularised problem,

$$F_\gamma^* \leq \langle C, x_0^* \rangle + \gamma \text{KL}(x_0^*|z) = F_0^* + \gamma \text{KL}(x_0^*|z),$$

hence $F_\gamma^* - F_0^* \leq \gamma \text{KL}(x_0^*|z)$, proving (11).

For the final bound, specialise to $z = \alpha \mathbf{1}$ with $\alpha := X_0^*/d$. Write $s := \|x_0^*\|_1 \leq X_0^*$. Expanding the divergence gives

$$\text{KL}(x_0^*|\alpha \mathbf{1}) = \sum_{i=1}^d \left(x_{0,i}^* \log \frac{x_{0,i}^*}{\alpha} - x_{0,i}^* + \alpha \right) \leq d\alpha + \left(s \log \frac{s}{\alpha} - s \right)_+ \leq d\alpha + s \log \frac{s}{\alpha} - s,$$

where we use $\sum_{i=1}^d x_{0,i}^* \log x_{0,i}^* \leq s \log s$ and $(t)_+ \geq t$ for all $t \in \mathbb{R}$, where $(t)_+ := \max\{t, 0\}$.

With $\alpha = X_0^*/d$ this yields

$$\text{KL}(x_0^*|(X_0^*/d)\mathbf{1}) \leq X_0^* + (s(\log d - 1))_+.$$

Finally, if $d \geq 3$ then $\log d - 1 \geq 0$, so the positive part can be dropped and, using $s \leq X_0^*$,

$$\text{KL}(x_0^*|(X_0^*/d)\mathbf{1}) \leq X_0^* + s(\log d - 1) \leq X_0^* + X_0^*(\log d - 1) = X_0^* \log d.$$

□

Proof of Theorem 3.2.

Proof. Split the total deviation into bias and optimisation pieces:

$$|F_0^* - F_\gamma(u^{(k)})| \leq \underbrace{|F_0^* - F_\gamma^*|}_{\text{bias}} + \underbrace{|F_\gamma^* - F_\gamma(u^{(k)})|}_{\text{optimisation}}.$$

By Lemma B.1, the existence of x_0^* with $\|x_0^*\|_1 \leq X_0^*$ implies $|F_0^* - F_\gamma^*| \leq \gamma X_0^* \log d$. With $\gamma = \varepsilon/(2X_0^* \log d)$ this gives $F_0^* - F_\gamma^* \leq \varepsilon/2$. The $O(1/k)$ dual rate of Theorem 3.1 yields $F_\gamma^* -$

$F_\gamma(u^{(k)}) \leq \frac{16X_\gamma U_\gamma^2 \|A\|_{1 \rightarrow 1}^2}{\gamma k}$. Our choice of $k \geq 32X_\gamma U_\gamma^2 \|A\|_{1 \rightarrow 1}/(\gamma \varepsilon)$ ensures $F_\gamma^* - F_\gamma(u^{(k)}) \leq \varepsilon/2$. Summing the two contributions gives the claim. □

C Extension to general Bregman divergences

The analysis presented above for the KL divergence extends naturally to a broader class of Bregman divergences. We briefly summarize the main ingredients and results.

Bregman divergence. Let $\mathcal{X} \subset \mathbb{R}^d$ be a nonempty closed convex set with nonempty relative interior, and let $\phi : \text{ri}(\mathcal{X}) \rightarrow \mathbb{R}$ be a Legendre-type convex function (proper, lower-semicontinuous, essentially smooth and strictly convex on $\text{ri}(\mathcal{X})$). The Bregman divergence induced by ϕ is defined as

$$D_\phi(x|y) := \phi(x) - \phi(y) - \langle \nabla \phi(y), x - y \rangle, \quad x, y \in \text{ri}(\mathcal{X}).$$

When $\mathcal{X} = \mathbb{R}_+^d$ and $\phi(x) = \sum_{i=1}^d x_i(\log x_i - 1)$, we recover the (non-normalized) KL divergence.

658 **Bregman-regularized linear program.** Fix a reference point $z \in \text{ri}(\mathcal{X})$ and a temperature $\gamma > 0$.
659 The Bregman-regularized problem reads

$$\min_{x \in \mathcal{C}_1 \cap \mathcal{C}_2} \langle C, x \rangle + \gamma D_\phi(x|z), \quad (\tilde{\mathcal{P}}_\gamma)$$

660 which can be equivalently written as a Bregman projection onto $\mathcal{C}_1 \cap \mathcal{C}_2$ of the shifted reference
661 $z_\gamma := \nabla \phi^*(\nabla \phi(z) - \frac{C}{\gamma})$.

662 **Dual problem.** Define the shifted dual reference $y_\gamma := \nabla \phi(z) - \frac{C}{\gamma}$. The dual objective is

$$\tilde{F}_\gamma(u) := \langle b, u \rangle - \gamma \phi^*\left(y_\gamma + \frac{A^\top u}{\gamma}\right),$$

663 with the primal-dual relation $x(u) = \nabla \phi^*(y_\gamma + \frac{A^\top u}{\gamma})$. The dual stationarity condition remains
664 $A x(u^*) = b$.

665 **Generalized Pinsker condition.** The key assumption replacing the Pinsker inequality is that there
666 exists a primal confinement set $\mathcal{X}_\gamma \subset \text{ri}(\mathcal{X})$, a norm $\|\cdot\|_*$ on $\mathbb{R}^d$, and a constant $\eta_\gamma > 0$ such that

$$\forall (x, y) \in \mathcal{X}_\gamma \times \mathcal{X}_\gamma, \quad D_\phi(x|y) \geq \eta_\gamma \|x - y\|_*^2. \quad (\tilde{\mathcal{C}}_\gamma)$$

667 For the KL divergence with $\mathcal{X}_\gamma = \{x \in \mathbb{R}_{++}^d : \|x\|_1 \leq X_\gamma\}$ and $\|\cdot\|_* = \|\cdot\|_1$, this holds with
668 $\eta_\gamma = 1/(2X_\gamma)$.

669 **Dual convergence result.** With the generalized Pinsker condition $(\tilde{\mathcal{C}}_\gamma)$ in place, the dual con-
670 vergence result (Theorem 3.1) extends as follows. Let $\|A\|_{* \rightarrow \mathcal{V}^*}$ denote the operator norm
671 from $(\mathbb{R}^d, \|\cdot\|_*)$ to the dual of the block-quotient norm. Assume the dual iterates satisfy
672 $\sup_k \|u^{(k)}\|_V \leq U_\gamma$ and that the primal iterates remain in $\mathcal{X}_\gamma$. Then

$$0 \leq \Delta_k \leq \frac{2 \|A\|_{* \rightarrow \mathcal{V}^*}^2}{\gamma \eta_\gamma} U_\gamma^2 \frac{1}{k}.$$

673 In the entropic case, $\eta_\gamma = 1/(2X_\gamma)$ and $\|A\|_{* \rightarrow \mathcal{V}^*} = \|A\|_{1 \rightarrow 1}$, recovering the bound of Theorem 3.1.

674 **Bias bound.** For the general Bregman case, the bias between the unregularized and regularized
675 optima satisfies

$$0 \leq F_\gamma^* - F_0^* \leq \gamma D_\phi(x_0^*|z),$$

676 where x_0^* is an optimal solution of the unregularized problem. Combined with the dual rate above,
677 this yields iteration complexity bounds analogous to Theorem 3.2.

678 **Remark C.1.** Beyond the entropic (KL) case, the Cressie–Read family of divergences provides
679 another concrete example. For $\alpha \in (0, 1)$, the generator $\varphi_\alpha(s) = \frac{s^\alpha - \alpha s + (\alpha - 1)}{\alpha(\alpha - 1)}$ yields a separable
680 Bregman divergence on $\mathbb{R}_{++}^d$. Condition $(\tilde{\mathcal{C}}_\gamma)$ then holds with $\eta_\gamma = \frac{1}{2d} X_\gamma^{\alpha-2}$ and $\|\cdot\|_* = \|\cdot\|_1$.
681 Unlike the KL case ($\alpha = 1$), the curvature-based argument introduces a dimension factor $1/d$ in the
682 Pinsker constant.

683 C.1 Quantum optimal transport as a noncommutative Bregman projection

684 The general-Bregman framework naturally extends to entropic quantum optimal transport (QOT).
685 This appendix presents a self-contained formulation aligned with the blueprint developed in the
686 paper and clarifies what is already in place versus what remains open. In short, the variational and
687 geometric setup is clear, but obtaining explicit, problem-dependent bounds for the key blueprint
688 constants is still an open step. Matrix-valued and quantum variants of optimal transport have appeared
689 in control and signal-processing formulations [Ning et al., 2015], in the mean-field/classical limits
690 of quantum mechanics [Golse et al., 2016, Caglioti et al., 2020], and in quantum-information and
691 learning settings [Chakrabarti et al., 2019]. Entropic regularization for matrix-valued/quantum OT
692 was developed in [Peyré et al., 2019], which is the closest numerical precedent for the variational
693 problem considered here.

694 **Variational QOT problem.** Let $\mathbb{H}_N$ denote Hermitian matrices, identify $N = nm$, and write
695 $\text{Tr}_2 : \mathbb{H}_{nm} \rightarrow \mathbb{H}_n$ and $\text{Tr}_1 : \mathbb{H}_{nm} \rightarrow \mathbb{H}_m$ for the two partial traces. Given positive semidefinite
696 marginals $A \in \mathbb{H}_{n,+}$ and $B \in \mathbb{H}_{m,+}$ with equal trace, and a Hermitian cost $C \in \mathbb{H}_{nm}$, the
697 unregularized QOT problem is

$$W_C(A, B) := \min_{T \succeq 0} \{ \text{tr}(CT) : \text{Tr}_2(T) = A, \text{Tr}_1(T) = B \}.$$

698 Its dual maximizes $\text{tr}(FA) + \text{tr}(GB)$ under the Loewner constraint $F \otimes I_m + I_n \otimes G \leq C$.
699 The exact entropic regularization is obtained from the noncommutative entropy generator $\phi(T) =$
700 $\text{tr}(T \log T - T)$ and a reference $Z \succ 0$:

$$\min_{\text{Tr}_2(T)=A, \text{Tr}_1(T)=B, T \succ 0} \text{tr}(CT) + \gamma D_\phi(T|Z), \quad D_\phi(T|Z) = \text{tr}(T(\log T - \log Z) - T + Z).$$

701 Equivalently, with $K_\gamma = \exp(\log Z - C/\gamma)$, this is the Bregman projection of K_γ onto the intersec-
702 tion of the two affine marginal constraints.

703 **Dual and block maps.** With dual variables $(F, G) \in \mathbb{H}_n \times \mathbb{H}_m$, the regularized dual objective is

$$\mathcal{F}_\gamma(F, G) = \text{tr}(FA) + \text{tr}(GB) - \gamma \text{tr} \exp\left(\frac{F \otimes I_m + I_n \otimes G - C}{\gamma}\right)$$

704 up to the harmless reference-dependent shift when $Z \neq I$. The primal-dual relation is

$$T(F, G) = \exp\left(\frac{F \otimes I_m + I_n \otimes G - C}{\gamma}\right),$$

705 or, for general Z , the same formula with $-C/\gamma$ replaced by $\log Z - C/\gamma$. Alternating maximization
706 defines exact block maps $\Phi_1(G) = \arg \max_F \mathcal{F}_\gamma(F, G)$ and $\Phi_2(F) = \arg \max_G \mathcal{F}_\gamma(F, G)$. These
707 maps are implicit in general, because matrix logarithms and exponentials do not commute; this lack
708 of a closed form is not itself fatal for the blueprint, which only requires exact block solves and
709 quantitative control of the resulting sweep.

710 **Quantum Pinsker geometry.** The correct analogue of the classical Pinsker inequality is the
711 quantum Pinsker inequality, historically attributed to Hiai–Ohya–Tsukada [Hiai et al., 1981]; it also
712 follows from Uhlmann’s earlier relative-entropy interpolation results [Uhlmann, 1977], and is stated
713 in modern textbook form in [Watrous, 2018]. For trace-one $P, Q \succeq 0$,

$$D(P\|Q) := \text{tr}(P(\log P - \log Q)) \geq \frac{1}{2} \|P - Q\|_1^2, \quad \|X\|_1 := \text{tr}|X|.$$

714 Thus the natural primal norm in the QOT blueprint is the trace/nuclear norm, and the natural
715 dual norm is the operator norm. For the pair of dual variables one should quotient by the gauge
716 $(F, G) \sim (F + cI_n, G - cI_m)$, leading to the spectral-variation seminorm

$$\|(F, G)\|_{\text{var, op}} := \inf_{c \in \mathbb{R}} \max\{\|F + cI_n\|_{\text{op}}, \|G - cI_m\|_{\text{op}}\}.$$

717 This is the direct noncommutative counterpart of the variation seminorm that appears for classical
718 Sinkhorn and graph flow-Sinkhorn.

719 **Open constants for the blueprint.** The variational dictionary is now clear: QOT fits the general-
720 Bregman proof with the nuclear norm on primal perturbations, the quotient operator norm on dual
721 variables, and a Pinsker constant supplied by quantum Pinsker on trace slices. What remains open
722 is to bound the problem-dependent constants needed to apply the generic convergence theorem. In
723 particular, one needs a usable bound on the quantum analogue of H_γ , equivalently a lower spectral
724 bound on the optimal T_γ , and a uniform quotient-operator bound on the dual iterates.

725 **Why the classical monotonicity proof does not transfer.** In the scalar Sinkhorn setting, mono-
726 tonicity plus translation equivariance of the dual maps implies non-expansiveness in the variation
727 seminorm. In QOT the natural order is the Loewner order, but this order is not a lattice order, and the
728 exact block maps Φ_1, Φ_2 are not operator monotone in the sense needed to reproduce the topical-map
729 argument. Preliminary numerical and differential evidence indicates that an individual block map
730 need not be non-expansive in the spectral-variation seminorm, although it remains open whether the
731 full sweep could satisfy a weaker orbit-confinement or non-expansiveness estimate. New strategies
732 are therefore needed, for instance via derivative projection-strip estimates, scalar-shifted positive
733 decompositions, or direct strong-concavity/cocoercivity bounds in the quotient operator geometry.

D Proofs for primal and dual iterate bounds

Bounding Primal and Dual Iterates. Our main result, Theorem 3.1, crucially relies on uniform primal and dual bounds, denoted respectively by $X_{\max}$ and $U_{\max}$. We first show in Subsection D that an $O(1/\gamma)$ bound on the primal variables automatically follows from a dual bound (in some cases, such as classical optimal transport, an $O(1)$ primal bound holds a priori, and this step can be skipped). Subsection D then presents a general strategy to establish dual boundedness, based on the assumption that the sweep map Ψ of the algorithm is non-expansive in the V -norm. This approach applies both to the classical optimal transport setting and to the optimal transport on graphs problem studied here.

Bounding Primal Iterates from Dual Iterates. In many cases, one directly has access to a primal bound X_γ (for instance, in classical OT, $X_\gamma = 1$). If this is not the case, the following proposition shows that it is always possible to derive a bound X_γ from U_γ , with the issue being that it blows as $X_\gamma \sim \|b\|_1 U_\gamma / \gamma$ when $\gamma \rightarrow 0$. The main workload is thus to bound the dual potential in the block quotient semi-norm, which needs to be done on a case-by-case basis and exploit the structure of the split (A_1, A_2) .

Proof of Proposition 4.2.

Proof. Recall that $F_\gamma(u) = \langle b, u \rangle - \gamma \|x(u)\|_1$, hence $\|x^{(k)}\|_1 = \frac{\langle b, u^{(k)} \rangle - F_\gamma(u^{(k)})}{\gamma}$. Because $F_\gamma(u^{(k)})$ is nondecreasing along the ascent, we have $F_\gamma(u^{(k)}) \geq F_\gamma(u^{(0)}) = F_\gamma(0)$. Therefore,

$$\|x^{(k)}\|_1 \leq \frac{\langle b, u^{(k)} \rangle - F_\gamma(0)}{\gamma}. \quad (13)$$

We first bound $\langle b, u^{(k)} \rangle$ using the block-quotient control. Choose $h^{(k)} = (h_1^{(k)}, h_2^{(k)}) \in \ker(A^\top)$ such that $\|u^{(k)} + h^{(k)}\|_\infty \leq \|u^{(k)}\|_V \leq U_\gamma$. Since $b \in \text{range}(A)$, one has $\langle b, h^{(k)} \rangle = 0$, hence

$$\langle b, u^{(k)} \rangle = \langle b, u^{(k)} + h^{(k)} \rangle \leq \|b\|_1 \|u^{(k)} + h^{(k)}\|_\infty \leq \|b\|_1 U_\gamma.$$

It remains to bound $-F_\gamma(0)$ explicitly. Since $u^{(0)} = 0$, $F_\gamma(0) = -\gamma \sum_{i=1}^d \exp(-\frac{C_i}{\gamma})$. Using $C_i \geq C_{\min}$ gives $\exp(-C_i/\gamma) \leq \exp(-C_{\min}/\gamma)$, hence $F_\gamma(0) \geq -\gamma d e^{-C_{\min}/\gamma}$. Plugging the two estimates into (13) yields

$$\|x^{(k)}\|_1 \leq \frac{\|b\|_1 U_\gamma + \gamma d e^{-C_{\min}/\gamma}}{\gamma} = \frac{\|b\|_1 U_\gamma}{\gamma} + d e^{-C_{\min}/\gamma},$$

which is the mass bound stated in Proposition 4.2. $\square$

Bounding Dual Iterates using Non-expansiveness of Ψ . The most difficult part is to control the dual iterates. In this section, we provide a generic blueprint that leverages the non-expansiveness of Ψ . It requires a primal bound H_γ (measuring deviation from the reference z in dual coordinates) and a control of the conditioning of the splitting through a decomposition constant κ .

Definition D.1 (Primal bound H_γ in dual coordinates). *We define $H_\gamma \in [0, +\infty]$ so that the minimizer x_γ of $(\mathcal{P}_\gamma)$ satisfies*

$$\|\log x_\gamma - \log z\|_\infty \leq H_\gamma. \quad (14)$$

Definition D.2 (Decomposition constant κ). *The decomposition constant is*

$$\kappa = \kappa(A_1, A_2) := \sup_{y \in \text{range}(A^\top), y \neq 0} \inf \left\{ \frac{\|w_1\|_\infty}{\|y\|_\infty} : \exists w_2 \text{ s.t. } A_1^\top w_1 + A_2^\top w_2 = y \right\} \in [0, +\infty]. \quad (15)$$

Proof of Proposition 4.1.

Proof. Let $u_\gamma = (u_{\gamma,1}, u_{\gamma,2})$ be any maximizer of (D_γ) . Since u_γ is globally optimal, each block is optimal given the other, hence $u_{\gamma,1}$ is a fixed point of Ψ : $u_{\gamma,1} = \Psi(u_{\gamma,1})$. Triangular inequality with $\|\cdot\|_{V_1}$ gives

$$\|u_1^{(k)}\|_{V_1} \leq \|u_1^{(k)} - u_{\gamma,1}\|_{V_1} + \|u_{\gamma,1}\|_{V_1}. \quad (16)$$

768 Using the iteration $u_1^{(k)} = \Psi^k(u_1^{(0)})$, the fixed-point property, and the non-expansiveness assumption,
 769 and triangular inequality

$$\|u_1^{(k)} - u_{\gamma,1}\|_{V_1} = \|\Psi^k(u_1^{(0)}) - \Psi^k(u_{\gamma,1})\|_{V_1} \leq \|u_1^{(0)} - u_{\gamma,1}\|_{V_1} \leq \|u_1^{(0)}\|_{V_1} + \|u_{\gamma,1}\|_{V_1}.$$

770 Combining with (16) gives

$$\|u_1^{(k)}\|_{V_1} \leq \|u_1^{(0)}\|_{V_1} + 2\|u_{\gamma,1}\|_{V_1}. \quad (17)$$

771 Let $y_\gamma := A^\top u_\gamma \in \text{range}(A^\top)$. By the definition of κ in (15), there exist w_1, w_2 such that
 772 $A_1^\top w_1 + A_2^\top w_2 = y_\gamma$ and

$$\|w_1\|_\infty \leq \kappa \|y_\gamma\|_\infty. \quad (18)$$

773 Set $h_1 := w_1 - u_{\gamma,1}$ and $h_2 := w_2 - u_{\gamma,2}$. Then $(h_1, h_2) \in \ker(A^\top)$, hence by definition of $\|\cdot\|_{V_1}$,

$$\|u_{\gamma,1}\|_{V_1} \leq \|u_{\gamma,1} + h_1\|_\infty = \|w_1\|_\infty \leq \kappa \|y_\gamma\|_\infty. \quad (19)$$

774 It remains to bound $\|y_\gamma\|_\infty$. At a dual maximizer u_γ , stationarity gives $Ax_\gamma = b$ with $x_\gamma = x(u_\gamma)$.
 775 Moreover, by definition of $x(u)$, componentwise, $A^\top u_\gamma = C + \gamma(\log x_\gamma - \log z)$. Using (14), we
 776 have $\|\log x_\gamma - \log z\|_\infty \leq H_\gamma$, hence

$$\|A^\top u_\gamma\|_\infty \leq \|C\|_\infty + \gamma \|\log x_\gamma - \log z\|_\infty \leq \|C\|_\infty + \gamma H_\gamma.$$

777 Inserting this into (19) yields

$$\|u_{\gamma,1}\|_{V_1} \leq \kappa (\|C\|_\infty + \gamma H_\gamma).$$

778 Combining with (17) gives the bound stated in Proposition 4.1. $\square$

779 E Balanced optimal transport specialization

780 This section does not present new contributions, it revisits the classical entropic approximation
 781 of discrete optimal transport (OT). This “warm-up” illustrates how the constants in Theorem 3.1
 782 specialise to familiar quantities and recovers the standard $O(C_{\max}/k)$ Sinkhorn rate. The next
 783 subsection extends the methodology to the graph-based W_1 distance (Section 5).

784 **Problem set-up.** For OT, the primal vector is a transport plan $x = P \in \mathbb{R}_{++}^{m_1 \times m_2}$, so that the
 785 dimension is $d = m_1 m_2$. The two marginals $b_1 \in \mathbb{R}_{++}^{m_1}$ and $b_2 \in \mathbb{R}_{++}^{m_2}$ are probability vectors,
 786 $\sum_i b_{1i} = \sum_j b_{2j} = 1$. A coupling $P \in \mathbb{R}_{++}^{m_1 \times m_2}$ must satisfy

$$A_1(P) = b_1, \quad A_2(P) = b_2, \quad \text{where} \quad A_1(P) := P \mathbf{1}_{m_2}, \quad A_2(P) := P^\top \mathbf{1}_{m_1}.$$

787 With a cost matrix $C \in \mathbb{R}_+^{m_1 \times m_2}$ the unregularised OT instance reads

$$W_C(b_1, b_2) := \min_{P \geq 0, A(P)=b} \langle C, P \rangle. \quad (20)$$

Entropic dual and Sinkhorn updates. The primal-dual relation between solution u of $(\mathcal{D}_\gamma)$ and $P(u)$ of $(\mathcal{P}_\gamma)$ with reference vector $z = b_1 \otimes b_2 = (b_{1,i} b_{2,j})_{i,j}$ (note that we use a separable reference measure for simplicity) reads

$$P(u)_{i,j} = b_{1,i} b_{2,j} e^{\frac{u_{1,i} + u_{2,j} - C_{i,j}}{\gamma}}.$$

788 The Sinkhorn update, written over the dual variable, corresponds to the two block updates where
 789 Ψ_1, Ψ_2 are soft-c-transforms

$$\Psi_1(u_2)_i := -\gamma \log \sum_j e^{\frac{-C_{i,j} + u_{2,j}}{\gamma}} b_{2,j}, \quad \Psi_2(u_1)_j := -\gamma \log \sum_i e^{\frac{-C_{i,j} + u_{1,i}}{\gamma}} b_{1,i}. \quad (21)$$

The block-quotient semi-norm is the so-called variation distance defined in (25):

$$\|\cdot\|_{V_1} = \|\cdot\|_{V_2} = \|\cdot\|_{\text{Var}}.$$

790 The following propositions give the value for the primal bound H_γ of Definition 4.1 and the decom-
 791 position constant κ defined in (4), which are the constants involved in the convergence rates. Recall
 792 that for OT, the reference measure is $z = b_1 \otimes b_2 = (b_{1,i} b_{2,j})_{i,j}$.

Proposition E.1 (H_γ for Sinkhorn). Assume $\min(b) > 0$ and for simplicity $C_{i,j} \geq 0$. Then, for γ small enough, one can take H_γ in Definition 4.1 as

$$H_\gamma = |\log(\min(b))| + \frac{2\|C\|_\infty}{\gamma}.$$

Proof. Let P_γ be the unique minimizer of the entropic OT problem and recall that $z_{i,j} = (b_1)_i(b_2)_j$. We need to bound $|\log(P_\gamma)_{i,j} - \log z_{i,j}|$ uniformly over all (i, j) .

Upper bound. From the optimality conditions, $P_\gamma = z \odot \exp((f \oplus g - C)/\gamma)$ where f, g are the Sinkhorn potentials satisfying the marginal constraints. The row marginal gives

$$\sum_j (b_2)_j \exp\left(\frac{f_i + g_j - C_{i,j}}{\gamma}\right) = 1.$$

Since all terms are non-negative and sum to 1, each term is at most 1: $(b_2)_j \exp\left(\frac{f_i + g_j - C_{i,j}}{\gamma}\right) \leq 1$, hence

$$\log(P_\gamma)_{i,j} - \log z_{i,j} = \frac{f_i + g_j - C_{i,j}}{\gamma} \leq -\log(b_2)_j \leq |\log(\min(b))|.$$

The same bound follows from the column marginal using $(b_1)_i$.

Lower bound. The optimal coupling reads $P_\gamma = \text{diag}(u)K \text{diag}(v)$ for scaling vectors u, v and $K_{i,j} := e^{-C_{i,j}/\gamma}$. Let $K_{\min} := \min_{a,b} K_{a,b} = e^{-\|C\|_\infty/\gamma}$ (using $\min C = 0$). From $P_\gamma = \text{diag}(u)K \text{diag}(v)$ and the marginal constraints, $(b_1)_i = u_i(Kv)_i$, $(b_2)_j = v_j(K^\top u)_j$, hence

$$(P_\gamma)_{i,j} = u_i K_{i,j} v_j = \frac{(b_1)_i (b_2)_j K_{i,j}}{(Kv)_i (K^\top u)_j}. \quad (22)$$

Since $K \leq \mathbb{1}\mathbb{1}^\top$ entrywise, $(Kv)_i \leq \sum_\ell v_\ell$ and $(K^\top u)_j \leq \sum_k u_k$. Also, the total mass constraint gives

$$1 = \sum_{a,b} (P_\gamma)_{a,b} = u^\top Kv \geq K_{\min} \left(\sum_k u_k \right) \left(\sum_\ell v_\ell \right),$$

so $(\sum_k u_k)(\sum_\ell v_\ell) \leq 1/K_{\min}$ and therefore $(Kv)_i (K^\top u)_j \leq \frac{1}{K_{\min}}$. Plugging this into (22) yields $(P_\gamma)_{i,j} \geq (b_1)_i (b_2)_j K_{i,j} K_{\min} \geq z_{i,j} K_{\min}^2$, hence

$$\log(P_\gamma)_{i,j} - \log z_{i,j} \geq 2 \log K_{\min} = -\frac{2\|C\|_\infty}{\gamma}.$$

Combining both bounds gives $|\log P_\gamma - \log z|_\infty \leq |\log(\min(b))| + \frac{2\|C\|_\infty}{\gamma}$. $\square$

Proposition E.2 (κ for classical OT). For classical OT, one can take $\kappa = 1$ in (4).

Proof. Let $Y \in \text{range}(A^\top)$, $Y \neq 0$. Choose any representation $Y_{ij} = \alpha_i + \beta_j$. For any scalar $c \in \mathbb{R}$, define $\alpha^{(c)} := \alpha - c\mathbb{1}_{m_1}$, $\beta^{(c)} := \beta + c\mathbb{1}_{m_2}$, so that $\alpha_i^{(c)} + \beta_j^{(c)} = \alpha_i + \beta_j = Y_{ij}$ for all (i, j) , hence $Y = A_1^\top \alpha^{(c)} + A_2^\top \beta^{(c)}$ for every c . We now show that one can choose c so that $\|\alpha^{(c)}\|_\infty \leq \|Y\|_\infty$. Indeed,

$$\min_{c \in \mathbb{R}} \|\alpha - c\mathbb{1}_{m_1}\|_\infty = \frac{1}{2} \left(\max_i \alpha_i - \min_i \alpha_i \right).$$

Fix any column index $j_0 \in \{1, \dots, m_2\}$. Then for all i, i' ,

$$\alpha_i - \alpha_{i'} = (\alpha_i + \beta_{j_0}) - (\alpha_{i'} + \beta_{j_0}) = Y_{ij_0} - Y_{i'j_0}.$$

Hence $|\alpha_i - \alpha_{i'}| \leq 2\|Y\|_\infty$, which implies $\max_i \alpha_i - \min_i \alpha_i \leq 2\|Y\|_\infty$. Therefore, $\min_{c \in \mathbb{R}} \|\alpha - c\mathbb{1}_n\|_\infty \leq \|Y\|_\infty$. Choose c attaining (or arbitrarily approximating) this minimum and set $w_1 := \alpha^{(c)}$, $w_2 := \beta^{(c)}$. Then $A_1^\top w_1 + A_2^\top w_2 = Y$ and $\|w_1\|_\infty \leq \|Y\|_\infty$, so

$$\inf \left\{ \frac{\|w_1\|_\infty}{\|Y\|_\infty} : \exists w_2, A_1^\top w_1 + A_2^\top w_2 = Y \right\} \leq 1.$$

Taking the supremum over $Y \in \text{range}(A^\top) \setminus \{0\}$ gives $\kappa \leq 1$. $\square$

820 **Corollary E.1.** *For classical OT, assuming $u^{(0)} = 0$ for simplicity, one may take*

$$X_\gamma = 1, \quad U_\gamma = 6 \|C\|_\infty + 2\gamma |\log(\min(b))|.$$

821 *Proof.* We apply Proposition G.2 with $\Sigma = \text{Id}$ and Proposition G.3 with $\tau = -1$. Together,
822 these imply (via Proposition G.1) that the sweep map Ψ is non-expansive with respect to the
823 variation seminorm $\|\cdot\|_{V_1}$. Propositions E.2 and E.1 give the values of H_γ and κ to be used in
824 Proposition 4.1. $\square$

825 This bound U_γ derived using the non-expansiveness of Ψ is not sharp. As shown in Chizat et al.
826 [2020], a more direct argument exploiting the closed-form expression (21) for Ψ , together with
827 the Lipschitz dependence of the dual variables on the cost matrix C , yields the tighter estimate
828 $U_\gamma = \frac{\|C\|_\infty}{2}$.

829 Using Corollary E.1, Theorem 3.2 shows that Sinkhorn achieves ε -additive accuracy on the optimal
830 transport cost using $O(\frac{n^2}{\varepsilon^2} \|C\|_\infty^2 \log n)$ arithmetic operations. This matches exactly the complexity
831 bounds established in Altschuler et al. [2017], Chakrabarty and Khanna [2021], Dvurechensky et al.
832 [2018], Chizat et al. [2020].

833 F Proofs for the graph W_1 Sinkhorn-flow algorithm

834 This appendix contains the proofs associated with Section 5. We keep the notation of the main text:
835 E is the directed edge set, $p = |E|$, $x = (f, g) \in \mathbb{F}^2$, $\mathcal{C}_1$ is the divergence constraint, and $\mathcal{C}_2$ is the
836 equality constraint $f = g$.

837 The linear operators are $A_1(f, g) = f\mathbf{1}_n - g^\top \mathbf{1}_n$ and $A_2(f, g) = f - g$, and for dual variables
838 $(v, U) \in \mathbb{R}^n \times \mathbb{F}$ the adjoint action is

$$A^\top(v, U) = \left((v_i + U_{i,j})_{(i,j) \in E}, (-v_j - U_{i,j})_{(i,j) \in E} \right) \in \mathbb{F} \times \mathbb{F}. \quad (23)$$

839 **Proof of Proposition 5.1.** For the projection onto $\mathcal{C}_1$, the KKT conditions for
840 $\min_{f,g} \text{KL}((f, g) || (h, h))$ subject to $-f\mathbf{1}_n + g^\top \mathbf{1}_n = b_1 - b_2$ give a multiplier $\lambda \in \mathbb{R}^n$
841 such that $\log(f_{i,j}/h_{i,j}) + \lambda_i = 0$ and $\log(g_{i,j}/h_{i,j}) - \lambda_j = 0$. Setting $s = e^{-\lambda}$ gives
842 $f = \text{diag}(s)h$ and $g = h \text{diag}(s)^{-1}$. Inserting these expressions in the divergence constraint yields
843 $(s \odot s) \odot (h\mathbf{1}_n) + s \odot (b_1 - b_2) - h^\top \mathbf{1}_n = 0$, whose positive root is the function ϕ of Proposition 5.1.
844 The projection onto $\mathcal{C}_2$ is a diagonal KL projection, hence the geometric mean.

845 **Proof of Proposition 5.2.** Combining the two closed-form projections just proved with the
846 parametrization $f^{(k)} = \text{diag}(s^{(k)})z^C \text{diag}(1/s^{(k)})$ gives the scaling update of Proposition 5.2.
847 Passing to $v = 2\gamma \log s$ and writing the sums in log-sum-exp form gives the stable formula (9).

848 **Proposition F.1** (Closed forms for $\|\cdot\|_{V_1}$ and $\|\cdot\|_{V_2}$). *One has $\|\cdot\|_{V_1} = \|\cdot\|_{\text{Var}}$ and $\|\cdot\|_{V_2} = \|\cdot\|_{\text{Var}}$,
849 where $\|\cdot\|_{\text{Var}}$ is the variation semi-norm defined in (25).*

850 **Proposition F.2** (Signed structure of the graph-flow split). *For the lifted variables $x = (f, g)$, take
851 the diagonal signature $\Sigma = \text{diag}(+I_E, -I_E)$ on the two flow blocks and the translation parameter
852 $\tau = +1$. Then the signed constraint operators satisfy the hypotheses of the monotone-block and
853 translation-equivariant criteria of Appendix G. Consequently, the full sweep map $\Psi = \Psi_1 \circ \Psi_2$ is
854 non-expansive for the variation quotient norm on the v block.*

855 *Proof.* With the sign convention detailed below, multiplying the g block by -1 turns each column of
856 the two block moment maps into a nonnegative incidence contribution: the $v_i + U_{i,j}$ and $-v_j - U_{i,j}$
857 terms become monotone in the signed coordinates. Thus $A_s \Sigma$ is entrywise nonnegative for the
858 two block updates. Moreover the two signed incidence contributions cancel after adding a constant
859 to the vertex potential and the same constant to the edge multiplier, giving the paired-balance
860 identity of Proposition G.3 with $\tau = +1$. Proposition G.2 gives monotonicity of each block map
861 and Proposition G.3 gives translation equivariance of the sweep. Applying Proposition G.1 yields
862 non-expansiveness in the variation quotient norm. $\square$

863 **Proposition F.3** (Closed form and non-expansiveness of Ψ_2). *One has*

$$\Psi_2(v)_{i,j} = \frac{1}{2} (v_j - v_i), \quad \text{and} \quad \|\Psi_2(v)\|_{V_2} \leq \|v\|_{V_1} \quad \text{for all } v \in \mathbb{R}^n. \quad (24)$$

Proof of Proposition F.1. The kernel relation $(\delta v, \delta U) \in \ker(A^\top)$ is equivalent to $\delta v_i + \delta U_{i,j} = 0$ and $-\delta v_j - \delta U_{i,j} = 0$ for every $(i, j) \in E$. Since the graph is connected, this implies $\delta v = c\mathbf{1}_n$ and $\delta U = -c\mathbf{1}_E$. Therefore Definition 3.1 gives $\|v\|_{V_1} = \inf_c \|v + c\mathbf{1}_n\|_\infty = \|v\|_{\text{Var}}$ and $\|U\|_{V_2} = \inf_c \|U - c\mathbf{1}_E\|_\infty = \|U\|_{\text{Var}}$.

Proof of Proposition F.3. Maximizing $F_\gamma(v, U)$ with respect to $U_{i,j}$ at fixed v gives $\Psi_2(v)_{i,j} = (v_j - v_i)/2$. Thus $\|\Psi_2(v)\|_{V_2} = \|\Psi_2(v)\|_{\text{Var}} \leq \|\Psi_2(v)\|_\infty \leq \frac{1}{2}(\max_i v_i - \min_i v_i) = \|v\|_{V_1}$.

Proposition F.4 (H_γ for flow Sinkhorn). Assume $0 < W_{\min} := \min_{(i,j) \in E} W_{i,j}$ and let $W_{\max} := \max_{(i,j) \in E} W_{i,j}$. Fix any feasible $\bar{f} \geq 0$ with $\bar{f}\mathbf{1} - \bar{f}^\top \mathbf{1} = b_1 - b_2$. One can take

$$H_\gamma = \log X_\gamma^* + \frac{2W_{\max}}{\gamma} + 3\|\log z\|_\infty, \quad X_\gamma^* := \frac{\langle W, \bar{f} \rangle + \gamma \text{KL}(\bar{f}|z)}{W_{\min}}.$$

Proof. The positive-cost bound of Lemma F.1 gives $\|f_\gamma\|_\infty \leq \|f_\gamma\|_1 \leq X_\gamma^*$, hence $\log(f_\gamma)_{i,j} - \log z_{i,j} \leq \log X_\gamma^* + \|\log z\|_\infty$. At optimality, the opposite orientations satisfy $(f_\gamma)_{i,j}(f_\gamma)_{j,i} = z_{i,j}z_{j,i} \exp[-(W_{i,j} + W_{j,i})/\gamma]$. Combining this identity with the same upper bound on $(f_\gamma)_{j,i}$ yields a lower bound on $(f_\gamma)_{i,j}$, and hence the displayed value of H_γ . $\square$

Lemma F.1 (Primal ℓ^1 bound under positive costs). Assume $C_i \geq C_{\min} > 0$ for all coordinates. If x_γ^* solves $(\mathcal{P}_\gamma)$, then for every feasible $\bar{x} \geq 0$ with $A\bar{x} = b$,

$$\|x_\gamma^*\|_1 \leq X_\gamma^* := \frac{\langle C, \bar{x} \rangle + \gamma \text{KL}(\bar{x}|z)}{C_{\min}}.$$

Proof. By optimality of x_γ^* and nonnegativity of KL, $C_{\min}\|x_\gamma^*\|_1 \leq \langle C, x_\gamma^* \rangle \leq \langle C, x_\gamma^* \rangle + \gamma \text{KL}(x_\gamma^*|z) \leq \langle C, \bar{x} \rangle + \gamma \text{KL}(\bar{x}|z)$. $\square$

Proposition F.5 (Graph- W_1 decomposition and iterate bounds). Let $\text{diameter}(E)$ be the maximum shortest-path distance between two vertices. Then $\kappa \leq 2 \text{diameter}(E)$. If $u^{(0)} = 0$, one may take

$$U_\gamma = 4 \text{diameter}(E)(W_{\max} + \gamma H_\gamma), \quad X_\gamma = \frac{\|b\|_1 U_\gamma}{\gamma} + p e^{-W_{\min}/\gamma}.$$

Proof. For $y = A^\top(v, U)$, write $y_{i,j}^{(f)} = v_i + U_{i,j}$ and $y_{i,j}^{(g)} = -v_j - U_{i,j}$. The edge field $g_{i,j} = y_{i,j}^{(f)} + y_{i,j}^{(g)} = v_i - v_j$ satisfies $\|g\|_\infty \leq 2\|y\|_\infty$. Fixing a root and integrating this gradient along shortest paths gives a representative $\tilde{v}$ with $\|\tilde{v}\|_\infty \leq 2 \text{diameter}(E)\|y\|_\infty$; defining $\tilde{U}_{i,j} = y_{i,j}^{(f)} - \tilde{v}_i$ gives $y = A^\top(\tilde{v}, \tilde{U})$. This proves the bound on κ . Proposition F.2 gives the non-expansiveness hypothesis required by Proposition 4.1; combining it with the bound on κ and Proposition F.4 gives the stated U_γ . Finally Proposition 4.2 gives the displayed X_γ . $\square$

Proof of Theorem 5.1. Choose a spanning tree and route the signed measure $q = b_1 - b_2$ along this tree: for each edge of the tree, the flow is the total signed mass of one component after cutting that edge. This produces a feasible flow $\bar{f}$ with total transported mass at most $\text{diameter}(E)\|q\|_1/2$ and cost at most $W_{\max}\text{diameter}(E)\|q\|_1/2$. Therefore the unregularized optimum has an explicit feasible mass bound $X_0^* = O(\text{diameter}(E))$ for probability inputs. Lemma F.1, Proposition F.4, and Proposition F.5 give explicit H_γ , $U_\gamma = O(\text{diameter}(E))$, and $X_\gamma = O(\text{diameter}(E)/\gamma + p e^{-W_{\min}/\gamma})$. With $\gamma \asymp \varepsilon$ and $p = o(1/\log(1/\varepsilon))$, the exponential term is lower order. Theorem 3.2 then requires $O(X_\gamma U_\gamma^2/\varepsilon^2) = O(\text{diameter}(E)^3/\varepsilon^4)$ iterations up to logarithmic factors, and each iteration costs $O(p)$ sparse edge operations.

897 G Non-expansiveness in Variation Semi-norm

Topical maps and non-expansiveness. We first recall a classical result of so-called “topical maps” in the nonlinear Perron–Frobenius/max-plus theory Candrall and Tartar [1980]. This ensures the non-expansiveness of a map for the variation seminorm, which is ℓ^∞ norm quotiented by translation

$$\|v\|_{\text{Var}} := \inf_{c \in \mathbb{R}} \|v + c\mathbf{1}\|_\infty = \frac{1}{2} \text{osc}(v) \quad \text{where} \quad \text{osc}(v) = \max(v) - \min(v), \quad (25)$$

where $\mathbf{1}$ denotes the all-ones vector of the appropriate dimension. The remaining part of this section shows that this result can be applied to $T = \Psi$, the sweep dual mapping. This result is pivotal to show dual boundedness as exposed in Section D.

Proposition G.1 (Monotone, translation–equivariant maps are non–expansive in the V –seminorm). *Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ satisfy:*

1. Monotonicity: $x \leq y$ coordinatewise $\Rightarrow T(x) \leq T(y)$ coordinatewise.
2. Translation–equivariance: $T(x + c\mathbf{1}) = T(x) + c\mathbf{1}$ for all $x \in \mathbb{R}^n$ and $c \in \mathbb{R}$.

Then T is non–expansive for $\|\cdot\|_{\text{Var}}$:

$$\|T(x) - T(y)\|_{\text{Var}} \leq \|x - y\|_{\text{Var}} \quad \text{for all } x, y \in \mathbb{R}^n.$$

Proof. Fix $x, y \in \mathbb{R}^n$ and set $d := y - x \in \mathbb{R}^n$. Let $a := \min_{1 \leq i \leq n} d_i, b := \max_{1 \leq i \leq n} d_i$, so that $a \leq d_i \leq b$ for every i , i.e. coordinatewise, $x + a\mathbf{1} \leq y \leq x + b\mathbf{1}$. By monotonicity of T and translation equivariance this implies

$$T(x + a\mathbf{1}) \leq T(y) \leq T(x + b\mathbf{1}) \quad \Rightarrow \quad T(x) + a\mathbf{1} \leq T(y) \leq T(x) + b\mathbf{1},$$

hence for each coordinate i , $a \leq (T(y) - T(x))_i \leq b$. Therefore

$$\max_i (T(y) - T(x))_i \leq b, \quad \min_i (T(y) - T(x))_i \geq a,$$

and taking the oscillation gives

$$\text{osc}(T(y) - T(x)) \leq b - a = \max_i d_i - \min_i d_i = \text{osc}(d) = \text{osc}(y - x).$$

Dividing by 2 yields the desired non–expansiveness. $\square$

Monotonicity. We assume there exists a diagonal signature matrix $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_d)$ with $\sigma_i \in \{\pm 1\}$ such that

$$B_s := A_s \Sigma \quad \text{is entrywise nonnegative, for } s \in \{1, 2\}. \quad (26)$$

Equivalently, $A_s = B_s \Sigma$ with $B_s \geq 0$ componentwise.

Using Σ and (26), define a partial order $\preceq_\Sigma$ on $\mathbb{R}^{m_2}$ by

$$u_2 \preceq_\Sigma v_2 \iff \begin{cases} (B_2^\top u_2)_i \leq (B_2^\top v_2)_i & \text{for all } i \text{ with } \sigma_i = +1, \\ (B_2^\top u_2)_i \geq (B_2^\top v_2)_i & \text{for all } i \text{ with } \sigma_i = -1. \end{cases} \quad (27)$$

Proposition G.2 (Monotonicity of block updates). *Assume (26). Then*

- (i) (Anti-monotonicity of Ψ_1) *If $u_2 \preceq_\Sigma v_2$, then $\Psi_1(v_2) \leq \Psi_1(u_2)$.*
- (ii) (Anti-monotonicity of Ψ_2) *If $u_1 \leq v_1$, then $\Psi_2(v_1) \preceq_\Sigma \Psi_2(u_1)$.*
- (iii) (Monotonicity of Ψ) *If $u_1 \leq v_1$ then $\Psi(u_1) \leq \Psi(v_1)$.*

Proof. For $u \in \mathbb{R}^m$, the primal-dual relation reads $x_i(u) = z_i^C \exp((A^\top u)_i)$. Introducing the moment maps M_s , the first-order optimality conditions read

$$M_s(\Psi_s(u)) = b_s \quad \text{where} \quad M_s(u) := A_s x(u) \in \mathbb{R}^{m_s}, \quad (28)$$

Using (26), write $A_s = B_s \Sigma$ with $B_s \geq 0$ componentwise. Then for each i ,

$$x_i(u) = z_i^C \exp\left(\frac{\sigma_i}{\gamma} ((B_1^\top u_1)_i + (B_2^\top u_2)_i)\right), \quad (29)$$

(i) *Anti-monotonicity of Ψ_1 .* Assume $u_2 \preceq_\Sigma v_2$. Let $u_1 := \Psi_1(u_2), v_1 := \Psi_1(v_2)$. Then $M_1(u_1, u_2) = b_1$ and $M_1(v_1, v_2) = b_1$ by (28). By Lemma G.1, for any fixed w ,

$$M_1(w, u_2) \leq M_1(w, v_2) \quad (\text{componentwise}). \quad (30)$$

928 In particular, $M_1(v_1, u_2) \leq M_1(v_1, v_2) = b_1$. Suppose for contradiction that $v_1 \not\leq u_1$, i.e. there
 929 exists an index j with $(v_1)_j > (u_1)_j$. Since $u_1 \mapsto M_1(u_1, u_2)$ is componentwise nondecreasing by
 930 Lemma G.1, we obtain

$$M_1(v_1, u_2) \geq M_1(u_1, u_2) = b_1.$$

931 Together with $M_1(v_1, u_2) \leq b_1$, this yields a contradiction. Hence $v_1 \leq u_1$, i.e. $\Psi_1(v_2) \leq \Psi_1(u_2)$.

932 (ii) *Anti-monotonicity of Ψ_2 in the signed order.* Assume $u_1 \leq v_1$. Let

$$u_2 := \Psi_2(u_1), \quad v_2 := \Psi_2(v_1).$$

933 Then $M_2(u_1, u_2) = b_2$ and $M_2(v_1, v_2) = b_2$ by (28). By Lemma G.1, for any fixed w ,
 934 $M_2(u_1, w) \leq M_2(v_1, w)$ (componentwise). In particular, $M_2(u_1, v_2) \leq M_2(v_1, v_2) = b_2$. Now
 935 suppose for contradiction that $v_2 \not\leq_\Sigma u_2$. Since $u_2 \mapsto M_2(u_1, u_2)$ is nondecreasing in the signed
 936 order by Lemma G.1, the failure of $v_2 \leq_\Sigma u_2$ implies $M_2(u_1, v_2) \not\leq M_2(u_1, u_2) = b_2$ (and in
 937 fact $M_2(u_1, v_2) \geq b_2$ componentwise). This contradicts $M_2(u_1, v_2) \leq b_2$. Hence $v_2 \leq_\Sigma u_2$, i.e.
 938 $\Psi_2(v_1) \leq_\Sigma \Psi_2(u_1)$.

939 (iii) *Monotonicity of Ψ .* Assume $u_1 \leq v_1$. By (ii), $\Psi_2(v_1) \leq_\Sigma \Psi_2(u_1)$. Apply (i) with $u_2 = \Psi_2(v_1)$
 940 and $v_2 = \Psi_2(u_1)$ to obtain $\Psi_1(\Psi_2(u_1)) \leq \Psi_1(\Psi_2(v_1))$, i.e. $\Psi(u_1) \leq \Psi(v_1)$. $\square$

941 **Lemma G.1** (Monotonicity of moment maps). *Assume (26). Then:*

942 (1) *For any fixed u_2 , the maps $M_1(\cdot, u_2)$ and $M_2(\cdot, u_2)$ are componentwise nondecreasing.*

943 (2) *For any fixed u_1 , the maps $M_1(u_1, \cdot)$ and $M_2(u_1, \cdot)$ are nondecreasing with respect to $\leq_\Sigma$.*

944 *Proof.* (1) *monotonicity in u_1 .* Fix u_2 . If $u_1 \leq v_1$ componentwise, then $B_1^\top u_1 \leq B_1^\top v_1$ component-
 945 wise because $B_1 \geq 0$. Hence for each i :

- 946 • if $\sigma_i = +1$, then the exponent in (29) increases, so $x_i(u_1, u_2) \leq x_i(v_1, u_2)$ and $\sigma_i x_i(\cdot) B_{s,i}$
 947 increases (since $B_{s,i} \geq 0$);
- 948 • if $\sigma_i = -1$, then the exponent decreases, so $x_i(u_1, u_2) \geq x_i(v_1, u_2)$, and multiplying by $\sigma_i =$
 949 -1 reverses the inequality: $\sigma_i x_i(u_1, u_2) \leq \sigma_i x_i(v_1, u_2)$, hence again $\sigma_i x_i(\cdot) B_{s,i}$ increases
 950 componentwise.

951 Summing over i yields $M_s(u_1, u_2) \leq M_s(v_1, u_2)$ componentwise for $s \in \{1, 2\}$, proving (1) and
 952 (2).

953 (2) *monotonicity in u_2 for the signed order.* Fix u_1 . If $u_2 \leq_\Sigma v_2$, then by definition (27),

$$(B_2^\top u_2)_i \leq (B_2^\top v_2)_i \text{ when } \sigma_i = +1, \quad (B_2^\top u_2)_i \geq (B_2^\top v_2)_i \text{ when } \sigma_i = -1.$$

954 Equivalently,

$$\sigma_i (B_2^\top u_2)_i \leq \sigma_i (B_2^\top v_2)_i \quad \text{for all } i.$$

955 Therefore the exponent in (29) increases for every i , which implies $x_i(u_1, u_2) \leq x_i(u_1, v_2)$. Now
 956 consider the contributions $\sigma_i x_i(\cdot) B_{s,i}$:

- 957 • if $\sigma_i = +1$, increasing x_i increases the nonnegative vector $x_i B_{s,i}$;
- 958 • if $\sigma_i = -1$, increasing x_i decreases the nonpositive vector $-x_i B_{s,i}$, i.e. increases it component-
 959 wise.

960 Summing over i yields $M_s(u_1, u_2) \leq M_s(u_1, v_2)$ componentwise for $s \in \{1, 2\}$, which is precisely
 961 nondecreasingness with respect to $\leq_\Sigma$. This proves (3) and (4). $\square$

962 **Translation equivariance.** Fix a sign parameter $\tau \in \{+1, -1\}$. We say that the pair (A_1, A_2)
 963 satisfies the *signed paired-balance condition* (with sign τ) if

$$A_1^\top \mathbf{1}_{m_1} + \tau A_2^\top \mathbf{1}_{m_2} = 0 \in \mathbb{R}^d. \quad (31)$$

964 One has $\tau = -1$ for classical OT and $\tau = +1$ for the lifted W_1 flow formulation on graphs.

965 **Proposition G.3** (Translation equivariance under signed paired-balance). *Assume the signed paired-*
 966 *balance condition (31). Then for every $c \in \mathbb{R}$,*

$$\Psi_2(u_1 + c \mathbb{1}_{m_1}) = \Psi_2(u_1) + \tau c \mathbb{1}_{m_2}, \quad \Psi_1(u_2 + c \mathbb{1}_{m_2}) = \Psi_1(u_2) + \tau c \mathbb{1}_{m_1}. \quad (32)$$

967 *Consequently, the full sweep $\Psi = \Psi_1 \circ \Psi_2 : \mathbb{R}^{m_1} \rightarrow \mathbb{R}^{m_1}$ is translation-equivariant:*

$$\Psi(u_1 + c \mathbb{1}_{m_1}) = \Psi(u_1) + c \mathbb{1}_{m_1}. \quad (33)$$

968 *Proof.* Condition (31) implies

$$F_\gamma(u_1 + c \mathbb{1}_{m_1}, w) = F_\gamma(u_1, w - \tau c \mathbb{1}_{m_2}) + \langle b_1, c \mathbb{1}_{m_1} \rangle + \langle b_2, \tau c \mathbb{1}_{m_2} \rangle, \quad (34)$$

969 where the last two terms are constants independent of w . Hence, maximizing over w ,

$$\arg \max_w F_\gamma(u_1 + c \mathbb{1}_{m_1}, w) = \arg \max_w F_\gamma(u_1, w - \tau c \mathbb{1}_{m_2}).$$

970 which is the desired result for Ψ_2 , the proof for Ψ_1 being similar. $\square$

971 H Lean Formalization Guide

972 The Lean development is intended as an audit trail for the mathematical structure of the paper. Each
 973 theorem, proposition, lemma, and corollary appearing in the manuscript is assigned a stable Lean alias,
 974 and the aliases are checked against the compiled Lean theorem constants. The formalization covers the
 975 main convergence chain, the regularized approximation theorem, the OT and graph- W_1 instantiations,
 976 and the auxiliary ingredients used in the appendices, including KL bias bounds, quotient-seminorm
 977 estimates, non-expansiveness interfaces, and the non-normalised Pinsker reduction of Appendix A.3.
 978 These auxiliary components are deliberately stated in reusable finite-dimensional forms, so that they
 979 can be inspected independently of the particular flow-Sinkhorn application.

980 **How to navigate the Lean code.** The current umbrella entry point is `lean/FlowSinkhorn/`
 981 `KLProjection.lean`, with project root `lean/FlowSinkhorn.lean`. For paper-oriented read-
 982 ing, the facade `lean/FlowSinkhorn/Paper.lean` imports section and appendix modules fol-
 983 lowing the manuscript structure. The canonical synchronization layer is `lean/FlowSinkhorn/`
 984 `KLProjection/StatementMap.lean`; it is re-exported by `lean/FlowSinkhorn/Paper/`
 985 `StatementMap.lean`. In this map, each paper-facing name, such as `thm_3_1`, `lem_A_1`, or
 986 `prop_F_5`, is an alias for one canonical Lean theorem constant, and each alias carries an
 987 implementation-file comment indicating where the proof is defined. The map itself is intentionally
 988 proof-free: it is a stable index from manuscript statements to proof-producing Lean modules.

989 The implementation modules are organized by mathematical role rather than by LaTeX order. The
 990 main groups are duality and primal-dual identities, dual convergence and rate estimates, primal/dual
 991 uniform bounds, finite-dimensional variation geometry, and the OT and graph- W_1 application layers.
 992 This organization keeps reusable proof infrastructure separate from the paper-facing alias layer while
 993 still allowing a reader to start from a statement label and jump directly to the corresponding proof file.

994 **Certification scope.** The certified development currently contains 26,596 non-comment, non-blank
 995 lines of Lean code in the KL-projection namespace, with 1,511 theorem/lemma declarations and
 996 36 direct definition/structure declarations under the repository audit counter. The paper-facing
 997 map is checked by scripts that verify alias completeness, compiled endpoint existence, statement
 998 numbering, and implementation-file locations. The build is green for `cd lean && lake build`
 999 `FlowSinkhorn.KLProjection.StatementMap`, and the KL-projection development contains no
 1000 `sorry`, `admit`, or local axiom declarations.

1001 A few examples illustrate what is certified. The formalized Pinsker appendix constructs the finite
 1002 sign selector, relates it to the ℓ^1 distance, builds the two-point Bernoulli measure used for the
 1003 sign test, invokes `mathlib`'s measure-theoretic Hoeffding lemma, normalizes the common-mass
 1004 variational inequality, performs the scalar quadratic optimization, and scales the result back to the
 1005 non-normalised form used in Appendix A.3. The per-step ascent formalization combines this Pinsker
 1006 layer with KL-gain certificates for the two block updates and then composes the two half-steps into
 1007 the full sweep inequality of Lemma A.1. The quotient-residual formalization proves the finite Holder

1008 step, the shifted-representative and gauge-orthogonality manipulations, and the conversion from
 1009 quotient-seminorm radii to the gap estimate used in Lemma A.2.

1010 The Lean formalization was carried out after the authors had completed and checked the LaTeX proofs.
 1011 It was assisted by ChatGPT (GPT-5.4) for proof generation and proof checking. The formalization
 1012 did not lead to significant changes in the authors’ mathematical arguments, but it did detect and help
 1013 correct arithmetic constant mistakes in intermediate bounds.

1014 I Notation

Notation	Meaning
d	Ambient dimension of the primal variable $x \in \mathbb{R}_+^d$.
$A = (A_1; A_2)$	Constraint matrix split into two blocks.
$b = (b_1; b_2)$	Right-hand side split compatibly with A_1, A_2 .
$C \in \mathbb{R}^d$	Linear cost vector in the unregularized linear program.
$z \in \mathbb{R}_{++}^d$	Positive reference measure/vector for the KL penalty.
$\gamma > 0$	Entropic regularization parameter.
$\mathcal{C}_1, \mathcal{C}_2$	Affine constraint blocks $\{A_1x = b_1\}$ and $\{A_2x = b_2\}$.
$\text{KL}(x z)$	Non-normalized Kullback–Leibler divergence from x to z .
z^C	Gibbs reference vector, $z_i^C = z_i \exp(-C_i/\gamma)$.
$F_\gamma(u)$	Dual objective associated with the entropically regularized problem.
$u = (u_1, u_2)$	Dual variable split according to the two constraint blocks.
$x(u)$	Primal variable recovered from a dual variable by the primal–dual relation.
Ψ_1, Ψ_2	Exact dual block maximization maps.
$\Psi = \Psi_1 \circ \Psi_2$	Full dual sweep map on the first dual block.
$\ \cdot\ _{V_1}, \ \cdot\ _{V_2}$	Block quotient seminorms induced by $\ker(A^\top)$.
$\ \cdot\ _V$	Maximum of the two block quotient seminorms.
Δ_k	Dual suboptimality gap $F_\gamma^* - F_\gamma(u^{(k)})$.
X_γ	Uniform ℓ^1 bound on primal iterates.
U_γ	Uniform block-quotient bound on dual iterates.
H_γ	Uniform bound on $\ \log x_\gamma - \log z\ _\infty$ at the regularized optimum.
$\kappa(A_1, A_2)$	Decomposition constant converting control of $A^\top u$ into quotient control of dual blocks.
V, E	Vertex and edge sets of the graph used for graph W_1 .
n, p	Number of graph vertices and directed sparse edge entries, respectively.
W	Edge-length/cost matrix on the graph.
$\mathbb{F}$	Sparse nonnegative flow cone supported on graph edges.
f, g	Duplicated graph-flow variables used in the flow-Sinkhorn splitting.
v, U	Vertex and edge dual variables in the graph-flow formulation.
$\text{diameter}(E)$	Maximum shortest-path distance between graph vertices.
Σ, τ	Signature matrix and scalar balance parameter used for signed monotonicity and translation equivariance.
D_ϕ	General Bregman divergence generated by a convex function ϕ .
η_γ	Generalized Pinsker constant in the Bregman extension.